

RoPE HAS A SPECTRAL BUDGET

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Paper under double-blind review

ABSTRACT

RoPE gives each attention head a finite frequency budget, yet geometric spacing couples the covered log-frequency range to one particular allocation within it; writing $x_k = -\log \omega_k = a + Rz_k$ separates sampled support from interior allocation. Under the stated uniform separation prior, phase-invariant canonical correlations yield an exact effective-dimension identity: over separations up to 4K, 23 slow pairs—46 nominal dimensions—have block-whitened effective dimension 2.00. A paired three-seed intervention then holds sampled support and the training protocol fixed while moving only the interior frequencies; every seed improves at every tested out-of-distribution length, identifying allocation as consequential rather than a disguised base change. A convex surrogate has a unique continuous minimizer whose inverse CDF yields EVQ-COSH, a closed-form table with no learned positional parameters. We test the consequences of this design axis in from-scratch and continued training, low-rank adaptation, frozen checkpoints, and a bidirectional video transformer, showing that RoPE’s frequency budget is not determined by range alone: allocation is a separate, controllable axis.

1 INTRODUCTION

RoPE specifies how a frequency rotates a query or key; it does not specify how a finite attention head should allocate its frequencies. Each head receives exactly $K = d_{\text{rot}}/2$ rotary pairs. The familiar scalar base determines the whole table only after one additionally assumes geometric spacing, thereby coupling the spectral interval to a particular allocation rule. We separate these two choices by writing

$$x_k = -\log \omega_k = a + Rz_k, \quad z_0 = 0, \quad z_{K-1} = 1, \quad (1)$$

where (a, R) is sampled support and z is the normalised interior allocation. Standard geometric RoPE fixes $z_k = k/(K-1)$. Support says which frequencies are available; allocation says how a finite head spends them. We ask whether the second choice changes trained behaviour when the first is held fixed, then turn the answer into a closed-form construction.

This distinction matters because geometric spacing is highly redundant at the slow end. For $L=4096$, $b=5 \times 10^5$, and $K=64$, the 23 pairs with $\omega L \leq 1$ occupy 46 nominal dimensions yet have block-whitened Rényi-2 effective rank 2.00. These channels can still carry content, but as positional functions they converge toward the same two-dimensional subspace. A geometric table therefore spends more than one third of its rotary pairs on near-duplicate slow positional directions. This is the spectral-budget problem studied here.

Long-context methods can change realised phases, sampled support, or interior placement, often more than one at once. Section 3 classifies range transport, scalar-base grids, learned tables, and broader positional operators under this decomposition. The control below instead pins both endpoints and the log-span before moving the interior frequencies.

It can. In a paired three-seed control, anchored EVQ-COSH moves only 30 interior frequencies relative to a geometric allocation at the fixed training support. Every seed favours the non-uniform allocation at every evaluated OOD length (§2.1). The arms share endpoints, initialisation, token order, optimiser, and training budget, so no scalar base can reproduce the intervention. Interior allocation z is therefore a consequential training variable, not a disguised base change.

A mature-checkpoint control makes the same variable visible after pretraining. On a released 1.485B OLMo checkpoint at 16K, with support, attention amplitude, data, decoder, and weights fixed, replacing a same-support geometric allocation by the tested derived allocation raises nine-task

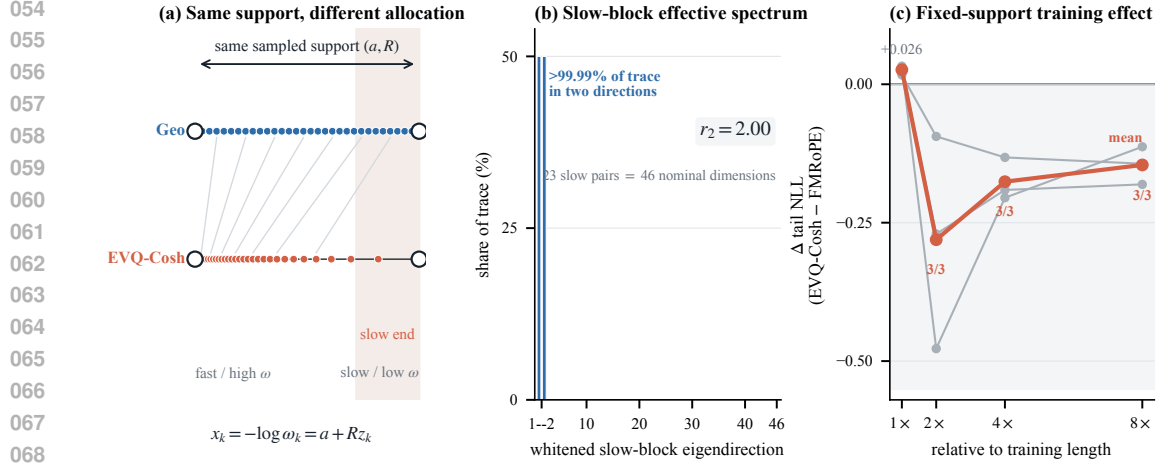


Figure 1: **One spectral range can encode different finite positional bases.** (a) At $K=32$, Geo and anchored EVQ-COSH use the same sampled endpoints and log-span but different interior allocation z . (b) For standard RoPE at $L=4096$, $b=5 \times 10^5$, and $K=64$, the 23 pairs with $\omega L \leq 1$ place more than 99.99% of their block-whitened trace in two eigendirections and have $r_2=2.00$ under the stated uniform separation prior. (c) The paired training intervention holds support fixed and moves only the 30 interiors; all three seeds improve at every tested OOD length.

RULER macro (20 rows/task) from 0.56% to 60.47%; the paired evaluation-row difference is 59.92 points with interval $[54.88, 64.80]$ (App. F.2). The common support already extends the Native slow endpoint by a factor of four, so the contrast is how its interior is spent. A coarse projected ramp reaches 61.04% and matches the derived profile within evaluation-row uncertainty. Two distinct reallocations recover the checkpoint, exposing the allocation axis directly; neither is the Cosh table used in the training experiments.

EVQ-COSH turns the identified degree of freedom into a deployable table. The unique minimizer of a stated convex surrogate has a Cosh density, and its inverse CDF gives every frequency in closed form. The construction adds no learned positional parameters and leaves the RoPE operator, architecture, and optimiser unchanged.

From-scratch MHA and MLA, full-parameter continuation, low-rank adaptation, frozen checkpoints, and a bidirectional 3D video DiT then track this design variable across the model-building lifecycle.

The geometry makes the finite-basis constraint exact; trained-model ordering still comes from controlled comparisons. Each rotary frequency contributes a two-dimensional sin/cos subspace, and Theorem 1 converts their mean phase-invariant redundancy into effective dimension while Proposition 2 characterises slow-frequency collapse. Table 1 then shows that higher static rank can accompany a catastrophic frozen mismatch, and Theorem 3 gives the exact limit of position-independent invertible Q/K compensation.

Contributions. (i) We identify interior allocation as a training variable with a paired three-seed intervention that fixes the sampled support and every scalar-base degree of freedom. (ii) We give a phase-invariant spectral-budget geometry for the full sin/cos basis: an exact effective-dimension identity, a slow-frequency collapse law, and a frozen-transplant obstruction. (iii) We derive EVQ-COSH as the closed-form solution of a stated convex surrogate and show the consequences of its zero-parameter allocation across architectures, training stages, scales, and pretrained checkpoints.

2 IDENTIFYING THE ALLOCATION AXIS

2.1 FIXED-SUPPORT TRAINING INTERVENTION

Q1: Does z change trained behaviour at fixed (a, R) ? Within each of three 151.9M training seeds, both arms share architecture, initialisation, token order, optimiser, LR schedule, global batch, a 499,974,144-token budget, and 32 evaluation anchors. The baseline is FMRoPE at fixed training support, implementing Oka et al. (2026b, §6.1) with $\theta=L_{\text{train}}=256$ and geometric exponents. Anchored EVQ-COSH ($\tau=4$) normalises its quantiles to the same extrema and log-span, so only the $K-2=30$ interior frequencies move while retaining the deployed Cosh table’s z . Every training seed favours anchored EVQ-COSH at every OOD length. The anchored EVQ-COSH minus FMRoPE NLL difference is +0.026 at the 256-token training length and $-0.281/-0.176/-0.146$ at 512/1K/2K (Fig. 1c). No geometric base can reproduce this intervention once the sampled support is pinned: *fixed support + different $z \Rightarrow$ different trained-model behaviour*.

Q2: Does this beat target-aware FMRoPE? No. Under target-matched support, anchored EVQ-COSH trails FMRoPE by +0.060/ +0.227/ +0.460 NLL at $2\times/4\times/8\times$, with FMRoPE favoured in all three seeds—0/3 for anchored EVQ-COSH (App. Fig. 4). The fixed-support result therefore identifies allocation z ; it does not establish an additive gain over target-aware support selection.

Q3: Does support selection exhaust the design? No. In a separate frozen-checkpoint protocol, support, attention amplitude, checkpoint, data, and decoder are fixed while only interior z changes; same-support geometric, derived, and coarse-ramp allocations score 0.56%/60.47%/61.04% on released-model nine-task 16K RULER (20 rows/task; App. F.2). This is neither the 151.9M training protocol nor a test of the Cosh table.

2.2 CROSS-CONFIGURATION AND SHAPE BREADTH

A pre-specified 50.9M factorial versus Geo spans two bases, two training lengths, three head dimensions, three paired seeds, scaled Cosh schedules, and a deformation-matched exponential; it supports the allocation axis without identifying a unique shape or strength (App. Table 9).

3 RELATED WORK

Write $x_k = -\log \omega_k = a + Rz_k$ as in Eq. (1). Prior work changes several, nonexclusive parts of this parameterisation: realised phases, support (a, R) , interior allocation z , or a larger positional operator. The coordinates are distinct but can interact.

Range and support. RoPE encodes relative displacement through rotation (Su et al., 2024); we take that operator as given and ask where its finite frequencies are placed. Position interpolation rescales positions, while YaRN and LongRoPE apply frequency-dependent or channel-wise range transport (Chen et al., 2023; Peng et al., 2024; Ding et al., 2024; Shang et al., 2025). The geometric-family base bounds attainable context length and governs a distinct extrapolation regime (Xu et al., 2024; Liu et al., 2024). FMRoPE moves (a, R) through that scalar base while retaining a uniform exponent grid (Oka et al., 2026b). MrRoPE gives a distinct training-free mixed-radix conversion of pretrained frequencies (Tian et al., 2026). These comparisons change support or realised phases; our controlled question pins both endpoints and the log-span before moving z .

Frequency use and operator scope. Trained RoPE uses its bands unevenly: query/key norms, data-induced dependency scales, and long-range retrieval expose different frequency structure (Barbero et al., 2025; Oka et al., 2026a; Wu et al., 2026; Chiang & Yogatama, 2025). HoPE, CoPE, and RoPE-ID respectively replace, clip, or restrict selected rotations (Chen et al., 2025; Li et al., 2026a; Wertheimer et al., 2026). Our geometry instead measures the phase-invariant effective dimension of all K two-dimensional pair subspaces, and the intervention reallocates those pairs without removing a plane. GRAPE generalises the planar group action and Selective RoPE learns input-dependent angles (Zhang et al., 2026; Movahedi et al., 2026); we retain the canonical planes and one session-static frequency table.

Learning and constructing frequency tables. LeRoPE learns one log-scale per frequency band, shared across layers and heads, and reports consistent non-geometric profiles on a 52M–2.5B model ladder (Karypis et al., 2026). In its 217M ablation, retraining with a frequency table frozen from an independent LeRoPE run retains 63.6% of the full validation-perplexity gain over RoPE, compared with 10.4% for partial RoPE. The frozen-table ablation locates substantial value in the table itself; our fixed-support intervention isolates the degree of freedom carrying that value. AdaRoPE learns per-head and per-dimension frequencies together with length-dependent attention scaling (Wang et al., 2026). Resonance RoPE analytically snaps wavelengths to close phase gaps during interpolation (Wang et al., 2024); EVQ-COSH instead obtains a shared fixed table from a stated convex surrogate and tests its interior allocation at fixed support. DAPE, FIRE, FoPE, and content-dependent repositioning learn a larger positional function than a fixed RoPE table (Zheng et al., 2024; Li et al., 2024; Hua et al., 2025; Li et al., 2026b).

4 THEORY

The central phenomenon is geometric: slow RoPE pairs collapse onto nearly the same positional subspace, so a finite table can spend many nominal dimensions without gaining comparable positional dimension. We first make this redundancy exact, then show why training binds the resulting basis to its weights, and finally derive a closed-form allocation that moves channels away from the collapse. Figure 1a–b visualizes the finite allocation problem; the analysis below makes it exact.

4.1 A FINITE TABLE IS A SET OF TWO-DIMENSIONAL SUBSPACES

Let $\phi \in [0, 1]$ parameterise log-frequency, $\omega(\phi) = b^{-\phi}$, let d_{rot} be the number of rotated dimensions and $K = d_{\text{rot}}/2$ the number of rotary pairs ($d_{\text{rot}} = d_{\text{head}}$ for standard MHA, smaller for compressed-RoPE architectures such as MLA).

We reserve z for the support-normalised coordinate in Eq. (1). Standard implementations use raw geometric exponents $u_k = k/K$, while deployed EVQ-COSH uses midpoint quantiles $u_k = (k + 1/2)/K$; endpoint-inclusive grids used for static diagnostics are labelled explicitly. Normalising the sampled geometric endpoints gives $z_k = k/(K - 1)$.

Equation (1) is a coordinate identity for any ordered positive table with nonzero span. In the geometric family $z_k = k/(K - 1)$, so fixing (a, R) fixes every channel. A second table with the same support and different interior z therefore cannot be represented by a different scalar base. Quantities such as median wavelength or the number of channels completing a cycle are summaries of z , not omitted scalar controls.

A rotary pair contributes $f_\omega(\Delta) = C \cos(\omega\Delta) + D \sin(\omega\Delta)$ to the relative-position attention logit, with C, D set by content. The object attached to ω is therefore the two-dimensional subspace

$$V_\omega = \text{span}\{\cos(\omega\Delta), \sin(\omega\Delta)\}, \quad (2)$$

not a cosine feature: an analysis that fixes one content phase sees a slice of V_ω and is not invariant to rotation inside the pair. Two channels are redundant to the extent that their subspaces are. With $x_\omega(\Delta) = [\cos(\omega\Delta) \ \sin(\omega\Delta)]$, Gram matrices $S_\omega = \mathbb{E}[x_\omega^\top x_\omega]$ and $H_{\omega\nu} = \mathbb{E}[x_\omega^\top x_\nu]$ over a separation prior, and the whitened cross-Gram $Q_{\omega\nu} = S_\omega^{-1/2} H_{\omega\nu} S_\nu^{-1/2}$ whenever $S_\omega, S_\nu \succ 0$, the singular values of $Q_{\omega\nu}$ are the canonical correlations between V_ω and V_ν , and

$$c_{\omega\nu} = \frac{1}{2} \|Q_{\omega\nu}\|_F^2 \in [0, 1] \quad (3)$$

is invariant to phase rotation inside a pair and to any change of basis of either subspace. For $\Delta \sim \text{Unif}[0, L]$, $H_{\omega\nu}$ has a closed form in $\sin t/t$ and $(1 - \cos t)/t$ at $t = (\omega \mp \nu)L$ (App. A).

4.2 THE SPECTRAL BUDGET IS AN EFFECTIVE-DIMENSION ACCOUNT

Block-whitening every pair gives a global correlation Gram R with diagonal blocks I_2 . Write $\bar{c}$ for the mean of the c_{ij} .

Theorem 1 (Spectral budget identity). *$\text{tr } R = 2K$ and $\text{tr}(R^2) = 2K[1 + (K-1)\bar{c}]$, so the Rényi-2 effective rank of the positional basis is exactly*

$$r_2(R) = \frac{(\text{tr } R)^2}{\text{tr}(R^2)} = \frac{2K}{1 + (K-1)\bar{c}}. \quad (4)$$

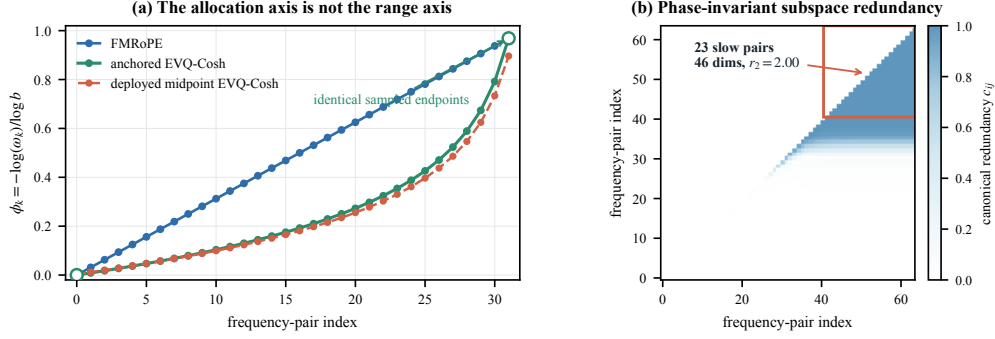


Figure 2: **Where a finite frequency table spends its budget.** (a) At $K=32$, Cosh concentrates interior channels toward the fast end while endpoint normalisation preserves sampled support; the deployed midpoint table is shown separately. (b) For standard RoPE at $L=4096$, $b=5 \times 10^5$, and $K=64$, phase-invariant full-subspace redundancy concentrates in the 23 pairs with $\omega L \leq 1$.

Table 1: **Weights co-adapt to their training-time frequency table.** Two seed-42 50M models trained with the geometric and EVQ-COSH tables, each evaluated under both runtime tables with all parameters frozen (1,920 head-query observations). The matched pairs remain near PPL 7, whereas both cross-swaps fail; the worst cell is also the one whose static r_2 improves most.

Trained weights	Runtime table	PPL	Rényi-2 rank r_2
Geo	Geo	7.14	4.57
Geo	EVQ-COSH	76.20	12.54
EVQ-COSH	Geo	23.05	4.57
EVQ-COSH	EVQ-COSH	7.16	12.54

The denominator turns the spectral budget into an exact Rényi-2 effective-dimension identity: larger mean pairwise redundancy lowers r_2 . Allocation changes the entire Gram spectrum, while block whitening removes per-channel energy and conditioning. The trained-model effect is then measured by the fixed-support intervention of §2.1.

The deflation is concentrated at the slow end.

Proposition 2 (Low-frequency collapse). *As $\omega L \rightarrow 0$, $V_\omega \rightarrow \text{span}\{1, \Delta\}$. For slow channels $x=\omega L$, $y=\nu L$, and $\epsilon = \max\{|x|, |y|\} \rightarrow 0$, $2 - \|Q_{x,y}\|_F^2 = \frac{19}{12600}(x^2 - y^2)^2 + O(\epsilon^6)$: slow subspaces approach a common two-dimensional limit at fourth order. Under the softmax metric for a fixed p with nondegenerate support on at least three distances, $F = \text{diag}(p) - pp^\top$ annihilates the constant direction, and the centred limit is $\text{span}\{\Delta - \mathbb{E}_p \Delta, \Delta^2 - \mathbb{E}_p \Delta^2\}$.*

Concretely, at $L=4096$, $b=5 \times 10^5$, $K=64$ the 23 pairs with $\omega L \leq 1$ occupy 46 nominal dimensions but have block-whitened $r_2 = 2.00$. Thus more than one third of the rotary pairs spend 46 nominal dimensions on approximately two effective positional dimensions. The channels may still carry content; the geometric redundancy tax is that many of them describe almost the same positional directions. Proposition 2 characterises the severe slow-band limit. The fixed-support identification also covers settings outside that asymptotic regime: in its $b=256$ control no sampled pair lies deep in the limit, and Theorem 1 measures overlap over the full table.

4.3 THE TABLE IS A TRAINING-TIME COORDINATE SYSTEM

Theorem 1 measures the geometry available to a table; trained behaviour additionally depends on the coefficients learned in that basis. To separate the two, we cross two trained 50M weight sets with two runtime tables at $L=512$, $b=5 \times 10^5$, and $\tau_{\text{EVQ}}=2.83$. All parameters remain frozen throughout this diagnostic (Table 1). Substituting EVQ-COSH for the geometric table at inference *raises* the block-whitened r_2 from 4.57 to 12.54 while perplexity degrades from 7.14 to 76.20.

Two consequences discipline the rest of the paper. First, collision, effective rank, and log-determinant describe the table without its learned coefficients; the task claims in §5 are therefore established on

trained models. Second, the two *self-consistent* systems are essentially tied in-window (7.14 vs. 7.16) while both mismatched cells collapse: on LM loss the table and weight main effects are +0.599 and -0.597, and their interaction contrast is -3.537 (configuration-level bootstrap CI [-5.17, -3.04]). The table behaves as a coordinate system fixed before training, with the weights learning coefficients in it.

Theorem 3 (Post-hoc transplant obstruction). *Let $R_\Omega(\Delta)$ be the block-rotation operator of a frequency multiset Ω . If position-independent invertible maps A, B satisfy $A^\top R_{\Omega'}(\Delta) B = R_\Omega(\Delta)$ for all Δ in an open interval containing zero, then Ω' and Ω have the same frequency multiset, up to sign and permutation. Repeated frequencies may mix within their full equal-frequency invariant subspace.*

The proof is short: $\Delta=0$ gives $A^\top B = I$, so the relation becomes a similarity $A^\top R_{\Omega'} A^{-\top} = R_\Omega$; differentiating at zero makes the generators similar, and similarity preserves the spectrum $\{\pm i\omega_k\}$ (App. A). The theorem therefore characterises exact function preservation by this broad class of fixed linear Q/K compensation. The finite-distribution protocols of §5.1 measure the complementary case in which Q/K or broader low-rank paths adapt to the new table.

4.4 AN ANALYTIC NON-UNIFORM ALLOCATION FAMILY

A positive density $\rho \in C^2([0, 1])$ with $\int_0^1 \rho = 1$ induces a discrete allocation by inverse-CDF quantisation, $\phi_k = F_\rho^{-1}(u_k)$; geometric RoPE is $\rho \equiv 1$. The exact canonical-correlation objective is oscillatory. To obtain a closed-form construction, we use the convex modelling surrogate

$$\mathcal{C}_{\text{app}}[\rho] = \frac{\alpha}{2} \int_0^1 \rho^2 d\phi + \frac{\beta}{2} \iint_{[0,1]^2} \rho(\phi) \rho(\psi) \min(\phi, \psi) d\phi d\psi, \quad (5)$$

with $\alpha, \beta > 0$. The diagonal term penalises channel-load concentration; $\min(\phi, \psi)$ is the Green kernel of $-\partial_\phi^2$ and encodes a penalty for pairs of channels that are both slow, matching the direction of Prop. 2. The full geometry and the surrogate play distinct roles: the former measures redundancy, while the latter supplies an analytic allocation family. The approximation is checked at the level needed for that construction: when the derived allocation is evaluated under the cosine-feature kernel in closed form, collision falls by 24–92% and its Shannon effective rank rises in all 12 tested configurations (App. A.5). The trained controls in §5 then test the learning consequences of the same surrogate-derived schedule.

Theorem 4 (Closed-form optimum under the surrogate). *Under (5) with $\alpha > 0, \beta \geq 0$ and constraints $\rho \in C^2([0, 1])$, $\rho > 0$, $\int_0^1 \rho = 1$, the unique constrained minimiser is*

$$\rho_\tau(\phi) = \frac{\tau \cosh(\tau(1-\phi))}{\sinh \tau}, \quad \tau = \sqrt{\beta/\alpha}; \quad (6)$$

at $\beta=0$ it is the $\tau \rightarrow 0$ limit $\rho_0 \equiv 1$.

Corollary 5 (Closed-form quantiles). *$F_\tau(\phi) = 1 - \sinh(\tau(1-\phi))/\sinh \tau$, so*

$$\phi_k(\tau) = 1 - \frac{1}{\tau} \operatorname{arcsinh}((1 - u_k) \sinh \tau), \quad u_k = \frac{k+1/2}{K}, \quad (7)$$

which tends to u_k as $\tau \rightarrow 0$, recovering the uniform-log midpoint grid.

The theorem selects the Cosh family exactly for (5); midpoint quantisation turns that density into a finite RoPE table. Because ρ_τ decreases with ϕ , it reallocates channels toward the fast end: at $K=64$ and $\tau=4$, 55 midpoint quantiles lie in the fast half of the log-frequency span. Cosh is therefore the exact analytic solution of the tractable surrogate, and the closed-form kernel diagnostic preserves the surrogate-derived improvement direction. Shape-matched trained controls below support non-uniform allocation without identifying a uniquely preferred curve.

Empirical zero-search reference. With d_{eff} the effective head dimension and Π the training protocol,

$$\tau = c(\Pi) d_{\text{eff}} / \sqrt{L_{\text{train}}}, \quad u_k = (k + \tfrac{1}{2})/K, \quad \omega_k^{\text{EVQ}} = b^{-\phi_k(\tau)}. \quad (8)$$

Set τ , evaluate (7), replace the inverse frequencies: no auxiliary loss, no extra parameters, no declared deployment length. A $d_{\text{eff}}/\sqrt{L_{\text{train}}}$ supplies an empirical reference family for standard full-RoPE text settings, with $d_{\text{eff}}=d_{\text{head}}$. We fix $c(\Pi)=1$ before training as a zero-search operating point and evaluate

it against Geo and neighbouring multipliers (App. C). Compressed rotary subspaces and bidirectional attention use pre-specified empirical τ values, testing the same allocation principle without invoking the full-RoPE rule. The deployed midpoint table has raw exponent positions $q_k = \phi_k(\tau)$. The exact-range diagnostic uses $z_k = (q_k - q_0)/(q_{K-1} - q_0)$ in Eq. (1). It therefore preserves the deployed table’s normalised Cosh shape exactly while replacing only its support embedding so that the sampled extrema match FMRoPE. This isolates z without changing the construction used by the complete deployed table.

5 CONSEQUENCES ACROSS MODEL-BUILDING REGIMES

The fixed-support experiment identifies the variable. The studies below follow its consequences through architectures, training stages, scales, and pretrained checkpoints.

5.1 ARCHITECTURE, RANGE COMPOSITION, AND FULL-PARAMETER SCALE

Scarce-channel MLA. Interior allocation remains consequential when the rotary budget is only $K=16$: a 432M MLA-style transformer (DeepSeek-AI, 2024) trains from scratch over three seeds with $d_{\text{rope}}=32$. Geo/EVQ-COSH PPL is 35.4/35.8 at the 8K training length. At 16K, however, it is 138.8/95.6: a 31.1% reduction, with every seed favouring EVQ-COSH (App. Table 13). This tested $K=16$ architecture makes every rotary pair scarce; the 16K gap locates a regime where allocation pays off.

Range composition. Holding the later index-space rule fixed reveals composition with the training-time allocation. In a 454M three-seed MHA study, a fixed YARN-STYLE rule raises 8K teacher-forced retrieval from 41% to 61% on Geo and from 53% to 100% on EVQ-COSH (App. Table 14). The code path, cut indices, and scaling formula are shared; because the underlying frequencies differ, this is a composition result: the same range operator acts on two learned spectral substrates.

Continuation and full-parameter scale evidence. The in-window/long-length crossover persists through full-parameter continuation and same-initialisation pretraining at 1.485B. Starting both 750M arms from a shared 2K Geo checkpoint and continuing all parameters for 500M tokens at 4K gives Geo/EVQ-COSH PPL 22.0/22.3 at 4K but 45.1/24.4 at 16K; strict autoregressive exact retrieval at 8K changes from 0% to 77.5% (App. Table 5). At 1.485B, a same-initialisation, same-scientific-recipe from-initialisation comparison with a 2.097B-token budget crosses in favour of EVQ-COSH at 8K and 16K; 122/128 and 126/128 held-out documents favour it at those lengths. Exact PPL values and protocol details appear in App. Fig. 5. Both larger protocols exhibit the same effective-context crossover: Geo retains the lower in-window PPL, while EVQ-COSH overtakes it at longer lengths. Together they carry the full-parameter scale evidence through 1.485B.

Frozen checkpoints. Frozen checkpoints isolate the same axis after pretraining. At fixed support, attention amplitude, checkpoint, data, and decoder, changing only interior z raises OLMo-2 unseen-nine 16K RULER macro (20 rows/task) from 0.56% to 60.47%; the paired evaluation-row difference is 59.92 points with interval [54.88, 64.80] on this fixed checkpoint and task set. A coarse projected ramp scores 61.04%; its paired evaluation-row interval against the derived profile includes zero under the same conditioning. The shared recovery ties the effect to the interior reallocation common to both profiles. On Qwen2.5 at 64K (2× Native), same-support geometric/derived is 57.75%/66.50%. These released-checkpoint interventions show that the allocation coordinate remains consequential after pretraining (App. F.2).

5.2 FROM EFFECTIVE CONTEXT TO REMOTE-CONTENT USE

1.485B task-family-adapted length transfer. Matched Q/K adaptation asks whether remote answer overlap moves with the table. Starting from protocol-matched Native and EVQ-COSH OLMo-2 Stage-A parents, both arms freeze inherited V/O LoRA and continue only Q/K for 300 steps with identical data, order, optimiser, trainable count, and 4K/8K/16K phase exposure; physical sequences remain at most 4K. On held-out 2WikiMultiHopQA, Native/EVQ-COSH token-F1 is 25.99/24.84% at 4K, 0.07/21.48% at 8K, and 0/8.57% at 16K (Ho et al., 2020; Bai et al., 2024).

Thus Q/K adaptation keeps the in-window QA score close to Native while only the EVQ-COSH arm carries substantial answer overlap to 8K in this task family. Training and test share task families but use disjoint rows, establishing task-family length transfer. An independent continuation over all 13 RULER families shows the same $2\times$ direction, with official macro 2.02/31.63% at 8K (Hsieh et al., 2024); all lengths and exact-match values appear in App. F.

8B causal remote-source use. At 8B, a source-block intervention tests whether long-context gains use the remote evidence rather than a local readout. On LLaMA-3-8B-Instruct (Grattafiori et al., 2024), rank-64 Q/K/V/O LoRA uses the same adaptation pipeline for Native and EVQ-COSH. PPL is 6.82/10.07, 108.96/24.07, and 991.48/127.91 at 8/16/32K, with the long-range direction holding on all 24 frozen natural-text packs and all three source domains. More importantly, at true 16K, median target-block hit@16 rises from 18.75% to 64.06%. Blocking answer-side decode attention to the remote gold block in every head, while preserving all prompt tokens and rotary KV indices, changes NLL by $-0.0095/+1.5055$. This position-preserving intervention shows causal dependence on the remote source in the tested adapted model rather than a local readout artifact. The task-family continuation is reported in App. F.

The allocation principle also appears under bidirectional 3D RoPE: a two-seed 129.6M video DiT favours EVQ-COSH on training, all-frame extrapolation, and far-frame MSE in both runs (App. B.5), extending the evidence to bidirectional 3D RoPE.

6 DISCUSSION

The main finding is that a RoPE base does not exhaust the design of a finite frequency table. Support (a, R) determines the sampled interval; allocation z determines how many channels are spent in each part of it. A scalar-base sweep explores only the geometric path $z_k = k/(K-1)$, whereas the fixed-support intervention moves outside that path without changing either endpoint. The resulting change in trained behaviour makes allocation a real design variable rather than an alternative notation for base selection.

Why the discovery matters. The spectral-budget identity reveals the pressure behind this variable. Slow geometric pairs collapse onto nearly the same positional subspace, so nominal rotary dimension can substantially overstate effective positional dimension. With $K=16$, the three-seed 432M MLA comparison shows that the construction remains effective in a compressed rotary subspace. The frozen OLMo comparison makes the axis especially visible: at the same support and amplitude, geometric allocation scores 0.56%, while derived and coarse-ramp reallocations both cross 60%. The geometric interior fails; two distinct reallocations recover.

The construction remains consequential across realistic model-building paths. At 454M, the same index-space YARN-STYLE rule has greater leverage on the EVQ-COSH substrate; at 750M and 1.485B, Geo leads in-window before EVQ-COSH overtakes it at longer lengths. Frozen interventions and low-rank adaptation extend the result to pretrained checkpoints. These protocols supply architecture, training-stage, and scale evidence around the fixed-support identification. At 8B, blocking answer-side attention to the distant source while preserving prompt tokens and rotary indices adds a causal source-use endpoint: the improved long-position probability depends on that source.

Static geometry and learned coefficients jointly determine how a frequency table behaves with fixed weights. The 50M crossing makes the interaction visible: a table with better static rank can still destroy perplexity. Theorem 3 characterises its exact limit—fixed invertible Q/K maps preserve all relative-position logits only when the frequency multisets agree. The frozen OLMo/Qwen controls address the practical finite-distribution case: changing z at fixed support and amplitude remains consequential after pretraining.

Support selection, interior allocation, and range transport act at different stages of the same spectrum. The fixed-support experiment isolates allocation; target-aware support and post-training range rules choose complementary degrees of freedom. LeRoPE shows that a learned table retains value when frozen for a new training run (Karypis et al., 2026). EVQ-COSH supplies such a table in closed form, and the paired intervention identifies what its allocation changes.

RoPE has a spectral budget because every head has finitely many rotary pairs. Geometric spacing is one way to spend that budget, not a neutral consequence of the rotary operator. Once support and allocation are separated, the latter can be analyzed and controlled as a consequential training-time RoPE design variable.

ETHICS STATEMENT

This work introduces no new dataset, deployed system, or human-subject component. The proposed training-time construction changes only the RoPE inverse-frequency table and adds no learned parameters; the mature-checkpoint studies use the frozen-table or LoRA interventions stated in their protocols. The table construction remains inexpensive to audit, ablate, or revert. All models, corpora, and evaluation suites used here are public research artefacts used within their stated licences.

A better allocated frequency table may reduce the adaptation or inference effort required to reach a target context range, and the same mechanism could be used to extend the context capability of language or video models, with the general risks associated with stronger generative models. Our evaluations measure positional and long-context endpoints, not factual reliability or safety; deployment claims require application-specific evaluation.

REPRODUCIBILITY STATEMENT

Equation (7) specifies the default midpoint EVQ-COSH table as a pure function of (τ, K) , with no learned parameters or auxiliary loss. Anchored support embeddings, endpoint $u_k=k/K$ quantisation in the mature OLMo protocols, and the inclusive static diagnostic are labelled where used; each retains the same continuous Cosh allocation law but defines a different finite grid or support. Equation (8) is the full-RoPE reference rule; every protocol reports its realised τ explicitly, including the separately fixed MLA and video settings and the mature-model tables in App. F. Complete proofs of Theorems 1, 3 and 4, Proposition 2, Corollary 5 and Lemma 7, together with every stated assumption, are in App. A. The identification protocol, including how sampled extrema and log-span are pinned, the exact pre-specified factorial design, and the corpora used by each arm, is in App. C; the mature-scale adaptation protocols, module budgets, phase-exposure mixtures, and evaluation harness settings are in App. F; remaining architectures and hyperparameters are in App. B. Seed scope is stated with every result. The anonymous supplement includes the exact-range training entriypoint and three-seed aggregate, the frozen figure generators, curated mature-protocol snapshots, and CPU tests, with a reviewer-facing file map and commands.

AI USE STATEMENT

We used generative AI tools to help develop conceptual frameworks and mathematical claims; propose and refine hypotheses; draft and check proofs; give feedback on methodology and experiments; implement methods and deterministic data-reformatting and analysis code; translate or rephrase research notes; and interpret results. We did not use generative AI to generate dataset content or to perform qualitative or thematic data analysis; survey, interview, and transcription tasks do not apply to this work.

For tasks for which disclosure is recommended, we also used these tools to suggest experimental parameters; write, edit, and test software and artefacts; create or modify figures; draft, edit, and structure parts of the manuscript; brainstorm and identify research gaps; search for and summarise literature; and format references. Every empirical quantity reported here comes from executed code over recorded run artefacts rather than language-model generation.

All AI-assisted work was reviewed before inclusion. We take responsibility for the final text, claims, proofs, code, figures, and experimental artefacts.

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A PROOF DETAILS

A.1 SUBSPACE GEOMETRY: CROSS-GRAM, BUDGET IDENTITY, AND COLLAPSE

Closed-form cross-Gram. For $\Delta \sim \text{Unif}[0, L]$ and $x_\omega(\Delta) = [\cos(\omega\Delta) \ \sin(\omega\Delta)]$, write $d = (\omega - \nu)L$, $s = (\omega + \nu)L$, $a(t) = \sin t/t$ and $b_*(t) = (1 - \cos t)/t$. Expanding the products of sinusoids and integrating termwise gives

$$H_{\omega\nu} = \mathbb{E}[x_\omega^\top x_\nu] = \frac{1}{2} \begin{bmatrix} a(d) + a(s) & b_*(s) - b_*(d) \\ b_*(s) + b_*(d) & a(d) - a(s) \end{bmatrix}, \quad (9)$$

and $S_\omega = H_{\omega\omega}$. Since $Q_{\omega\nu} = S_\omega^{-1/2} H_{\omega\nu} S_\nu^{-1/2}$ is the whitened cross-Gram of two two-dimensional subspaces, its singular values are the canonical correlations, and $c_{\omega\nu} = \frac{1}{2} \|Q_{\omega\nu}\|_F^2$ is invariant under $x_\omega \mapsto x_\omega O$ for any invertible O (in particular under rotation of the phase inside a pair), because S_ω transforms congruently and the whitening cancels O .

Basis invariance. The cancellation can be stated exactly. Under independent invertible basis changes $x_\omega \mapsto x_\omega A_\omega$ and $x_\nu \mapsto x_\nu A_\nu$, let $U_j = S_j^{1/2} A_j (A_j^\top S_j A_j)^{-1/2}$ for $j \in \{\omega, \nu\}$. Then $U_j^\top U_j = I$, and the transformed whitened cross-Gram is $\tilde{Q}_{\omega\nu} = U_\omega^\top Q_{\omega\nu} U_\nu$. Left and right orthogonal factors preserve the two canonical correlations and therefore $c_{\omega\nu}$. The metric is consequently attached to the positional subspaces themselves, not to a chosen sine/cosine phase convention.

Proof of Theorem 1. R is the block matrix with blocks $R_{ij} = Q_{\omega_i \omega_j}$ and $R_{ii} = I_2$. Hence $\text{tr } R = \sum_{i=1}^K \text{tr } I_2 = 2K$. For the second moment, $\text{tr}(R^2) = \sum_{i,j} \text{tr}(R_{ij} R_{ji}) = \sum_i \text{tr}(I_2) + \sum_{i \neq j} \|Q_{\omega_i \omega_j}\|_F^2 = 2K + 2K(K-1)\bar{c}$, using $R_{ji} = R_{ij}^\top$ and the definition $\bar{c} = \frac{1}{K(K-1)} \sum_{i \neq j} c_{\omega_i \omega_j}$ with $\|Q_{ij}\|_F^2 = 2c_{ij}$. Dividing gives (4). $\square$

The identity determines Rényi-2 rank exactly from the mean pairwise redundancy. Shannon effective rank and log-determinant retain higher-order multi-subspace dependence on the full spectrum.

Proof of Proposition 2. With $t = \Delta/L$ and $x = \omega L$, $\cos(xt) = 1 - x^2 t^2/2 + O(x^4)$ and $\sin(xt)/x = t - x^2 t^3/6 + O(x^4)$, so after rescaling the second coordinate the basis of V_ω tends to $\{1, \Delta\}$. Substituting these expansions into (9) and whitening, the leading deficit between two slow subspaces is $2 - \|Q_{x,y}\|_F^2 = \frac{19}{12600}(x^2 - y^2)^2 + O(\epsilon^6)$, where $\epsilon = \max\{|x|, |y|\} \rightarrow 0$; numerically at $x=0.05, y=0.10$ the exact-to-leading ratio is 1.00058. For the softmax metric with fixed p , $F = \text{diag}(p) - pp^\top$ satisfies $F\mathbf{1} = 0$, so constants are annihilated; centring by $\mathbb{E}_p$ and rescaling gives $\sin(\omega\Delta)/\omega \rightarrow \Delta - \mathbb{E}_p \Delta$ and $-2\cos(\omega\Delta)/\omega^2 \rightarrow \Delta^2 - \mathbb{E}_p \Delta^2$, which span the centred limit whenever p has nondegenerate support on at least three distances. $\square$

The reported $r_2=2.00$ for the 23 slow pairs uses the standard $u_k=k/K$ grid and the uniform separation prior above. Other priors change that static numerical Gram. The fixed-range trained-model contrast of §2.1 instead follows its paired empirical protocol.

A.2 EFFECTIVE DIMENSION AND MULTISCALE COVERAGE

The budget identity is an exact account of positional dimension under its separation prior. A second requirement is where that dimension is placed across scales. The distinction is already visible in an exact construction.

Lemma 6 (Full in-window rank can coexist with exact out-of-window aliasing). *Let $n_1, \dots, n_K$ be distinct positive integers and set $\omega_k = 2\pi n_k/L$. Under $\Delta \sim \text{Unif}[0, L]$, the block-whitened correlation Gram of the corresponding full sin/cos pairs is $R = I_{2K}$, hence $r_2(R) = 2K$. Nevertheless the complete positional feature map satisfies $\Phi(\Delta + L) = \Phi(\Delta)$ for every Δ .*

Proof. Fourier orthogonality on $[0, L]$ makes the sine and cosine functions at distinct integer harmonics mutually orthogonal, while each nonzero harmonic has self-Gram $\frac{1}{2}I_2$. Block whitening therefore produces the identity Gram and attains the full nominal rank. For every k , $e^{i\omega_k(\Delta+L)} = e^{i\omega_k\Delta} e^{i2\pi n_k} = e^{i\omega_k\Delta}$, proving the period- L alias for all pairs jointly. $\square$

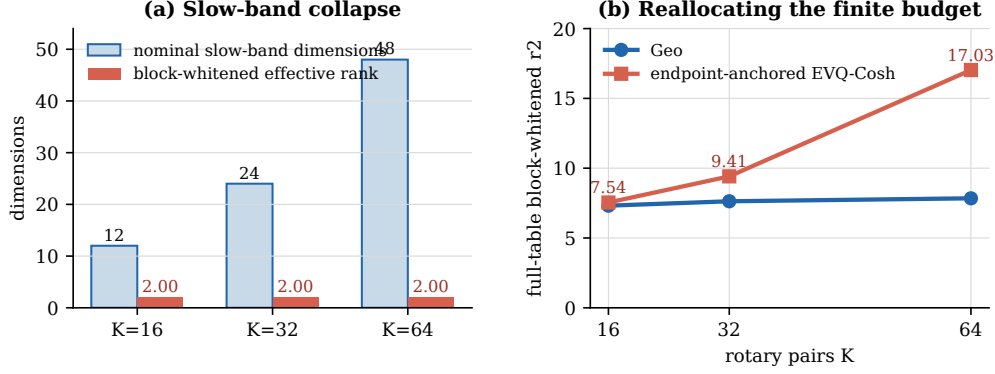


Figure 3: **Spectral-budget anatomy across channel counts.** Panel (a) uses the endpoint-inclusive diagnostic grid at $L=4096$ and $b=500K$: as K grows from 16 to 64, the nominal dimension assigned to pairs with $\omega L \leq 1$ grows from 12 to 48, while their phase-invariant effective rank remains approximately two. Panel (b) evaluates the complete fixed-endpoint table under the same uniform separation prior, using the zero-search reference $\tau=2K/\sqrt{L}$; anchored EVQ-COSH spends progressively more of the available full-pair dimension than Geo as the budget grows. These are static positional-basis diagnostics; the trained effect is identified separately in App. C.

Lemma 6 separates two design questions. Theorem 1 measures how many distinct positional directions a finite table supplies over the chosen interval; multiscale coverage is a separate criterion for how those directions span separations beyond one interval. The fixed-support experiments evaluate that allocation with trained models, while Table 1 shows why the learned coefficients must be kept with the basis that trained them.

A.3 POST-HOC TRANSPLANT OBSTRUCTION

Proof of Theorem 3. Both R_Ω and $R_{\Omega'}$ are block-diagonal with 2×2 rotation blocks, hence $R_\Omega(0) = R_{\Omega'}(0) = I$. Evaluating the hypothesis at $\Delta=0$ gives $A^\top B = I$, so $B = A^{-\top}$ and the hypothesis becomes the similarity $A^\top R_{\Omega'}(\Delta) A^{-\top} = R_\Omega(\Delta)$ on an interval. Each side is the exponential of a constant generator, $R_\Omega(\Delta) = \exp(\Delta G_\Omega)$ with $G_\Omega = \bigoplus_k \omega_k J$, $J = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. Differentiating at $\Delta=0$ gives $A^\top G_{\Omega'} A^{-\top} = G_\Omega$, so the generators are similar and share a spectrum. The spectrum of G_Ω is $\{\pm i\omega_k\}_{k \leq K}$ with multiplicity, so the frequency multisets agree up to sign and permutation. When a frequency is repeated, similarity may mix the entire equal-frequency invariant subspace; it need not preserve the original two-dimensional block decomposition. For integer positions the same argument is replaced by a one-step spectral argument: evaluating at $\Delta = 0$ again gives $B = A^{-\top}$, while $\Delta = 1$ makes $R_{\Omega'}(1)$ and $R_\Omega(1)$ similar. Their eigenvalue multisets $\{e^{\pm i\omega_k}\}$ therefore agree, which identifies frequencies up to sign, permutation, and the 2π alias. $\square$

This exact statement covers *fixed, position-independent, invertible* compensation. Approximate retraining on finite data, nonlinear maps, and changes to V/O or the residual path are separate intervention classes.

A.4 PROOF OF THE CLOSED-FORM VARIATIONAL OPTIMUM

This subsection proves Theorem 4 directly from the stated variational objective. Starting from the kernel

$$K_{\text{app}}(\phi, \psi) = \alpha \delta(\phi - \psi) + \beta \min(\phi, \psi), \quad (10)$$

the main-text collision functional is (5), which under the mass normalization $\int_0^1 \rho = 1$ with Lagrange multiplier ν has constrained first variation

$$\alpha \rho(\phi) + \beta g(\phi) + \nu = 0, \quad g(\phi) = \int_0^1 \rho(\psi) \min(\phi, \psi) d\psi. \quad (11)$$

Existence. For rigor, view $\mathcal{C}_{\text{app}}$ first as a functional on $L^2([0, 1])$ and minimize over $\mathcal{A} = \{\rho \in L^2([0, 1]) : \rho \geq 0 \text{ a.e., } \int_0^1 \rho = 1\}$. The $\alpha\|\rho\|_2^2$ term is coercive and weakly lower semicontinuous, the Green-kernel term is positive semidefinite and continuous, and the mass/nonnegativity constraints are weakly closed. A minimizer therefore exists by the direct method. Strict convexity gives uniqueness. The Euler solution derived below is strictly positive, so the nonnegativity constraint is inactive a posteriori and the classical C^2 derivation is justified.

Direct differentiation of g gives

$$g'(\phi) = \int_{\phi}^1 \rho(\psi) d\psi, \quad g''(\phi) = -\rho(\phi). \quad (12)$$

Differentiating the first variation twice in ϕ eliminates ν (a constant) and yields the homogeneous ODE

$$\rho''(\phi) - \tau^2 \rho(\phi) = 0, \quad \tau = \sqrt{\beta/\alpha}. \quad (13)$$

Evaluating the pre-differentiated stationarity together with $g(0)=0$, $g'(1)=0$, and $\int_0^1 \rho = 1$ gives the derivative boundary conditions

$$\rho'(0) = -\tau^2, \quad \rho'(1) = 0, \quad (14)$$

since differentiating $\alpha\rho(\phi) + \beta g(\phi) + \nu = 0$ once and evaluating at $\phi=1$ gives $\alpha\rho'(1) + \beta g'(1) = 0$ hence $\rho'(1)=0$, and at $\phi=0$ gives $\alpha\rho'(0) + \beta g'(0) = 0$ with $g'(0) = \int_0^1 \rho = 1$, so $\rho'(0) = -\beta/\alpha = -\tau^2$. Substituting $\rho(\phi) = C_1 \cosh(\tau\phi) + C_2 \sinh(\tau\phi)$ into (13) and (14) together with $\int_0^1 \rho = 1$ gives the unique positive interior stationary density

$$\rho_{\tau}(\phi) = \frac{\tau \cosh(\tau(1-\phi))}{\sinh \tau}, \quad (15)$$

which is main-text Theorem 4. The mass constraint $\int \rho=1$ is not lost: it enters the boundary condition at $\phi=0$ via $g'(0)=\int_0^1 \rho=1$. Positivity holds on the closed interval since $\rho_{\tau}(\phi) \geq \rho_{\tau}(1) = \tau/\sinh \tau > 0$ for $\tau > 0$ (with $\rho_{\tau}(0) = \tau \coth \tau > 0$); the inequality constraint $\rho > 0$ is therefore inactive throughout $[0, 1]$ and the unconstrained Euler–Lagrange solution coincides with the KKT-constrained solution. For the degenerate case $\beta=0$ the ODE reduces to $\rho''=0$ with $\rho'(0)=\rho'(1)=0$ and $\int \rho=1$, forcing $\rho \equiv 1$, which is also the $\tau \rightarrow 0$ limit of (15).

Convexity / PSD of $\mathcal{C}_{\text{app}}$. For $\alpha > 0$ and $\beta \geq 0$, the surrogate is convex because both quadratic terms are positive semidefinite. The diagonal $\alpha \int \rho^2$ is manifestly PSD on L^2 , and the Green-kernel cross term satisfies the identity

$$\iint_{[0,1]^2} f(\phi) f(\psi) \min(\phi, \psi) d\phi d\psi = \int_0^1 \left(\int_s^1 f(u) du \right)^2 ds \geq 0. \quad (16)$$

Thus the stationary density obtained above is the unique constrained *minimizer* (not merely a critical point), and the surrogate is well-defined as a quadratic functional on $L^2([0, 1]) \supset C^2([0, 1])$ with no further regularity assumption needed for the δ -component. Conditional on $\mathcal{C}_{\text{app}}$, this derivation is exact.

Why this branch. The homogeneous Cosh solution has an analytically invertible CDF, turning the continuous optimum into an explicit finite table. The functional test below checks the resulting allocation against the exact oscillatory kernel, and the trained comparisons supply the behavioural test.

A.5 SURROGATE QUALITY: FUNCTIONAL VALIDATION

The broadband surrogate $K_{\text{app}} = \alpha\delta + \beta \min$ supplies the closed-form density. We validate its allocation functionally against the closed-form cosine-feature kernel $K_{\cos}(\phi_1, \phi_2) = \int D(\Delta) \cos(\omega(\phi_1)\Delta) \cos(\omega(\phi_2)\Delta) d\Delta$: does the derived allocation also reduce the collision objective under K ? This directly tests whether the improvement direction survives the exact oscillatory kernel.

Table 2 answers affirmatively. For a uniform distance prior $D(\Delta) = 1/L$ on $[0, L]$, we compute $K_{\cos}$ at the discrete allocation points for Geo and EVQ-COSH, then evaluate the normalized collision score $C = \sum_{i < j} K_{ij}^2 / (K_{ii} K_{jj})$ and Shannon effective rank (the exponential of the eigenvalue distribution’s Shannon entropy). The inclusive grid $u_k = k/(K-1)$ exposes the complete diagnostic interval. This exact-kernel test complements the phase-invariant full sin/cos geometry of Theorem 1; trained midpoint tables are evaluated in §5. The 12 reported configurations comprise a six-length text sweep at fixed $b=500K$ and $K=32$, two additional text bases at $L=2048$, and four video bases at $L=32$ and $K=16$.

Table 2: Functional surrogate check on the inclusive $u_k=k/(K-1)$ diagnostic grid. EVQ-COSH reduces collision and increases Shannon effective rank under the closed-form cosine-feature kernel $K_{\cos}$ in all 12 tested configurations. Theorem 1 supplies the complementary phase-invariant full-subspace geometry; §5 evaluates trained midpoint tables.

Configuration	Collision C		Shannon rank	
	Geo	EVQ-COSH (Δ_C)	Geo	EVQ-COSH
<i>Text: $K=32, b=500K, \text{ sweep } L$</i>				
$L=128$	226.3	17.7 (−92%)	2.5	17.0
$L=256$	192.6	27.6 (−86%)	3.0	13.9
$L=512$	163.2	43.3 (−73%)	3.6	10.8
$L=1024$	133.9	59.0 (−56%)	4.4	8.8
$L=2048$	109.0	67.5 (−38%)	5.4	8.0
$L=4096$	87.0	66.3 (−24%)	6.6	8.1
<i>Text: $K=32, L=2048, \text{ sweep } b$</i>				
$b=10K$	34.8	19.2 (−45%)	13.0	17.3
$b=100K$	79.4	47.0 (−41%)	7.1	10.4
<i>Video: $K=16, L=32, \text{ sweep } b$</i>				
$b=100$	22.6	14.2 (−37%)	4.8	6.4
$b=1K$	42.3	28.0 (−34%)	2.8	3.9
$b=10K$	55.4	39.8 (−28%)	2.2	2.9
$b=50K$	63.3	47.2 (−26%)	2.0	2.5

EVQ-COSH reduces collision under $K_{\cos}$ in all 12 configurations (24%–92%), and Shannon effective rank increases in every row. In the fixed-base text sweep, the relative reduction grows as L shortens. The separate base sweeps remain positive but are not monotone in b .

A.6 SINGLE-CROSSING BUDGET SHIFT

Lemma 7 (Single-crossing budget shift). *For every $\tau > 0$, ρ_τ crosses the uniform density exactly once, at*

$$\phi_c(\tau) = 1 - \tau^{-1} \operatorname{arcosh}(\sinh \tau / \tau) \leq 1 - 1/\sqrt{3}. \quad (17)$$

Proof of Lemma 7. For $\tau > 0$, $\rho_\tau(\phi) = \tau \cosh(\tau(1-\phi)) / \sinh \tau$ is strictly decreasing on $[0, 1]$, with $\rho_\tau(0) = \tau \coth \tau > 1$ and $\rho_\tau(1) = \tau / \sinh \tau < 1$. Hence the crossing is unique, and solving $\rho_\tau(\phi_c) = 1$ gives (17). To bound it, expand

$$\frac{\sinh \tau}{\tau} = \sum_{n \geq 0} \frac{\tau^{2n}}{(2n+1)!} \geq \sum_{n \geq 0} \frac{\tau^{2n}}{3^n (2n)!} = \cosh\left(\frac{\tau}{\sqrt{3}}\right),$$

where the coefficient-wise inequality is $3^n \geq 2n+1$. Monotonicity of arcosh yields $\operatorname{arcosh}(\sinh \tau / \tau) \geq \tau / \sqrt{3}$, proving the bound. The three-region masses follow directly from $F_\tau(\phi) = 1 - \sinh(\tau(1-\phi)) / \sinh \tau$ at $\phi_L = \log(L/2\pi) / \log b$ and $\phi_M = \log(M/2\pi) / \log b$. $\square$

A.7 SURROGATE SELF-CONSISTENCY THEOREM

The surrogate $\mathcal{C}_{\text{app}}$ contains two integral terms $T_1(\tau) = \int_0^1 \rho^2 d\phi$ and $T_2(\tau) = \int_0^1 \int_0^1 \rho(\phi) \rho(\psi) \min(\phi, \psi) d\phi d\psi$, where $\rho(\phi) = \tau \cosh(\tau(1-\phi)) / \sinh \tau$ is the normalized cosh density.

Theorem 8 (Surrogate self-consistency). *For all $\tau > 0$:*

$$\tau^2 T_2(\tau) + T_1(\tau) = \tau \coth \tau. \quad (18)$$

Proof. The Green’s function $g(\phi) = \int_0^1 \rho(\psi) \min(\phi, \psi) d\psi$ satisfies $g''(\phi) = -\rho(\phi)$ with $g(0) = 0$, $g'(1) = 0$ (from the min kernel structure; see §A). Define $h(\phi) = \rho(0) - \rho(\phi)$. Then $h'' = -\rho''$, and under the pure-tether ODE $\rho'' = \tau^2 \rho$ this gives $h'' = -\tau^2 \rho$, with $h(0) = 0$ and $h'(0) = -\rho'(0)$.

Define $f(\phi) = \tau^2 g(\phi) - h(\phi)$. Then $f'' = \tau^2 g'' - h'' = -\tau^2 \rho - (-\tau^2 \rho) = 0$, so f is linear in ϕ . The boundary values give $f(0) = \tau^2 g(0) - h(0) = 0$ and $f'(0) = \tau^2 g'(0) + \rho'(0) = \tau^2 - \tau^2 = 0$ (using $g'(0) = \int_0^1 \rho = 1$ and $\rho'(0) = -\tau^2$ from the pure-tether density). Hence $f \equiv 0$, yielding $\tau^2 g(\phi) = h(\phi) = \rho(0) - \rho(\phi)$, i.e.,

$$g(\phi) = \frac{\rho(0) - \rho(\phi)}{\tau^2}. \quad (19)$$

Integrating against ρ :

$$T_2 = \int_0^1 \rho(\phi) g(\phi) d\phi = \frac{1}{\tau^2} [\rho(0) \underbrace{\int_0^1 \rho}_{=1} - T_1] = \frac{\rho(0) - T_1}{\tau^2}. \quad (20)$$

Since $\rho(0) = \tau \cosh \tau / \sinh \tau = \tau \coth \tau$, rearranging gives (18). $\square$

An immediate corollary follows.

Corollary 9 (Closed-form T_2).

$$T_2(\tau) = \frac{\sinh 2\tau - 2\tau}{4\tau \sinh^2 \tau}. \quad (21)$$

This follows by substituting $T_1(\tau) = \tau^2(1 + \sinh(2\tau)/(2\tau))/(2 \sinh^2 \tau)$ and $\rho(0) = \tau \coth \tau$ into the theorem.

The identity checks the two surrogate energy terms exactly; trained-loss evidence is supplied by §5.

A.8 FROM SCALING STRUCTURE TO AN OPERATING DEFAULT

The post-softmax model uses $\mathcal{F}(\tau) = \frac{1}{2} S_{\chi^2}(\tau) - \lambda U(\tau, L)$ with $S_{\chi^2}(\tau) = \tau^4/(45d_{\text{head}}) + O(\tau^6)$ and $U(\tau, L) = (d_{\text{head}}/L)[Q_0 + \tau^2 Q_1(L, b) + O(\tau^4)]$. Its leading stationary balance is

$$\tau_*^2 = 45\lambda Q_1(L, b) \frac{d_{\text{head}}^2}{L}. \quad (22)$$

Thus the model supplies the powers of d_{head} and L ; the finite coefficient $c(\Pi) = \sqrt{45\lambda Q_1}$ depends on the chosen utility, base, and protocol. Setting $c=1$ is the deployed zero-search convention fixed before training; the finite coefficient remains protocol-dependent.

Here

$$q(x) = \frac{1}{2} + \frac{\sin(2x)}{4x} - \left(\frac{\sin x}{x} \right)^2 = \text{Var}_{t \sim U[0,1]}[\cos(xt)]$$

and $Q_1(L, b) = \int_0^1 \eta(\phi) q(Lb^{-\phi}) d\phi$. This specified phase-variance utility supplies the scaling model; the exact oscillatory collision objective and trained-transformer loss are evaluated separately.

The finite- τ evidence evaluates a configuration-independent zero-search prior. In a staged 99-run study, the default beats the midpoint-discretised Geo baseline in 7/9 configuration means and 18/27 seed pairs. The evaluated neighbours use the discrete ratios 0.75 $\times$, 1.25 $\times$, or 1.5 $\times$ of the formula; the independent exact-range factorial tests 0.75 $\times$, 1.00 $\times$, and 1.25 $\times$. Their ordering varies across configurations. These tested multipliers sample the formula point and its registered neighbours. The evidence supports $c=1$ as a useful zero-search prior relative to that Geo baseline. The full run matrix and statistical units are recorded in `data/curated/phasel6_99run_manifest.csv`; the supplement includes the runner `scripts/core_text_phases/phasel6_formula_optimality_sweep.py`. This discrete design establishes the tested operating prior, not a continuous basin or per-configuration optimum.

A.9 POST-SOFTMAX OPERATING-POINT MODEL

The shape surrogate and the operating-point model answer different questions. The surrogate selects the Cosh family; the proposition below gives the scaling structure used to set its finite strength.

Proposition 10 (Semi-analytic operating-point scaling). *On the pure-tether family, assuming $Q_1(L, b) > 0$ and a diffuse post-softmax baseline $p_0=1/L$, the Pearson- χ^2 channel-load stiffness and the post-softmax transport utility admit small- τ expansions $S_{\chi^2}(\tau) = c_S \tau^4/d_{\text{head}} + O(\tau^6)$ with $c_S=1/45$, and $U(\tau, L) = (d_{\text{head}}/L)[Q_0 + \tau^2 Q_1(L, b) + O(\tau^4)]$, where $Q_1(L, b) = \int_0^1 \eta(\phi) q(Lb^{-\phi}) d\phi$, $\eta(\phi) = (1-\phi)^2/2 - 1/6$ and $q(x) = 1/2 + \sin(2x)/(4x) - (\sin x/x)^2$. The leading-order stationary point of $\mathcal{F}(\tau) = \frac{1}{2} S_{\chi^2}(\tau) - \lambda U(\tau, L)$ satisfies*

$$\tau_*^2 = 45\lambda Q_1(L, b) \cdot \frac{d_{\text{head}}^2}{L}, \quad \tau_* \propto \frac{d_{\text{head}}}{\sqrt{L}}.$$

The explicit $d_{\text{head}}/\sqrt{L}$ factor is modulated by $Q_1(L, b)$, so this is a scaling structure rather than a pure power law, and the prefactor $c(\Pi)=\sqrt{45\lambda Q_1(L, b)}$ remains convention- and protocol-dependent. The proposition supplies a small- τ scaling model; the deployed configurations near $\tau=4$ use the exact Cosh warp. The trained-model controls in App. C evaluate the independently fixed empirical default $c=1$.

A.10 CHANNEL-LOAD MOTIVATION FOR PEARSON χ^2

Normalization convention. Throughout this paper, $S_{\chi^2}(\tau)$ denotes the d_{head} -normalized stiffness $S_{\chi^2}(\tau) := \frac{1}{d_{\text{head}}} \int_0^1 (1-\rho_\tau)^2 / \rho_\tau d\phi$, with leading expansion $\tau^4/(45 d_{\text{head}}) + O(\tau^6)$ consistent with Proposition 10. Identities below that produce a bare $\tau^4/45$ leading coefficient apply to $d_{\text{head}} S_{\chi^2}(\tau)$ (the un-normalized integral); the d_{head} factor enters via the surrogate diagonal $\alpha=1/d_{\text{rot}}$ ($=1/d_{\text{head}}$ for MHA via §A.8; $=1/d_{\text{rot}}$ for compressed-RoPE MLA via §G.1).

Channel-load axioms. Inverse-CDF quantization places channels with cell width $\ell(\phi)=1/\rho(\phi)$ and samples channels according to $dP_\rho=\rho d\phi$. Under the modeling axioms (i) channel-measure averaging, (ii) load ℓ as the primitive local quantity, (iii) second central moment, (iv) uniform density has zero stiffness, (v) local additivity, the canonical load-variance penalty is

$$S(\rho) = \text{Var}_{\phi \sim \rho d\phi} \left[\frac{1}{\rho(\phi)} \right] = \int_0^1 \frac{(1-\rho(\phi))^2}{\rho(\phi)} d\phi = S_{\chi^2}(\rho). \quad (23)$$

Under the stated axioms, this is the canonical load-variance penalty.

A.11 DISCRETE-CHANNEL TRANSPORT GAP UNDER FINITE CHANNEL COUNTS

Setup. Inverse-CDF quantization replaces the continuous density ρ by an atomic measure. Let $F(\phi) = \int_0^\phi \rho$, $Q(u)=F^{-1}(u)$, midpoint grid $u_k=(k-\frac{1}{2})/K$, and $\phi_k=Q(u_k)$. The empirical channel measure is $\mu_K=K^{-1} \sum_{k=1}^K \delta_{\phi_k}$. Direct $\|\rho_K - \rho\|$ is ill-posed: one cannot subtract an atomic measure from a density. We therefore work in Wasserstein distance for transport and in a quantile-cell histogram ρ_K for density.

Transport bounds. Assume $0 < m \leq \rho \leq M$ and $\|\rho'\|_\infty \leq B$; then Q is $(1/m)$ -Lipschitz. The midpoint quantization errors satisfy

$$W_\infty(\mu_K, \mu_\rho) \leq \frac{1}{2Km}, \quad W_1(\mu_K, \mu_\rho) \leq \frac{1}{4Km}. \quad (24)$$

Defining the quantile-cell histogram $\rho_K(\phi)=1/(K|I_k|)$ on $I_k=[Q((k-1)/K), Q(k/K)]$, the histogram density satisfies

$$\|\rho_K - \rho\|_1 \leq \frac{B}{Km}, \quad \|\rho_K - \rho\|_\infty \leq \frac{B}{Km}. \quad (25)$$

The transport bounds couple each quantile cell to its midpoint; the histogram bound follows from the mean-value theorem and $|I_k| \leq 1/(Km)$.

Application to EVQ-Cosh. For $\rho_\tau(\phi) = \tau \cosh(\tau(1-\phi))/\sinh \tau$, $m_\tau = \tau/\sinh \tau$ and $B_\tau = \tau^2$. Substituting,

$$W_1(\mu_{K,\tau}, \mu_{\rho_\tau}) \leq \frac{\sinh \tau}{4K\tau}, \quad \|\rho_{K,\tau} - \rho_\tau\|_1 \leq \frac{\tau \sinh \tau}{K} = \frac{\tau^2}{K} + O\left(\frac{\tau^4}{K}\right). \quad (26)$$

Kernel-integrated errors. For a smooth two-variable kernel K_{sm} with Lipschitz constant L_K in each argument,

$$\left| \iint K_{\text{sm}} d\mu_K d\mu_K - \iint K_{\text{sm}} d\mu_\rho d\mu_\rho \right| \leq \frac{L_K}{2Km}. \quad (27)$$

The kernel bound applies the W_1 estimate to each argument. The singular δ -component of K_{app} is represented by its cell average.

High-resolution distortion. For midpoint scalar quantisation with a smooth weighted distortion w , the standard high-resolution expansion is

$$\mathcal{D}_K[\rho] = \frac{1}{12K^2} \int_0^1 \frac{w(\phi)}{\rho(\phi)^2} d\phi + O(K^{-3}). \quad (28)$$

This is the finite-channel mechanism: a shaped ρ can reduce the weighted inverse-density load $\int w/\rho^2$; the absolute benefit scales as K^{-2} for smooth quantization error and as K^{-1} for transport/kernel terms.

Dependence of the bounds on channel count. The standard MHA configuration has $d_{\text{head}}=d_{\text{rope}}=64$ and $K=32$; the MLA study has $d_{\text{head}}=64$, $d_{\text{rope}}=32$, $d_{\text{nope}}=32$ and $K=16$. Halving K scales the K^{-2} component of this distortion bound by $4\times$ and the K^{-1} component by $2\times$. The MLA experiment (§G.1) supplies the corresponding trained-model test at $K=16$.

Large- τ regime. At large τ , $m_\tau=\tau/\sinh \tau$ becomes small and the $1/(Km_\tau)$ factor grows. The transport bounds are informative while this factor remains controlled.

B EXPERIMENTAL DETAILS

B.1 SCOPE OF THIS APPENDIX

The exact-range and mature-model protocols are documented separately in Apps. C and F. This section records the submitted scratch-model and cross-modal experiments that remain supporting evidence for the substrate-range and cross-modal scope claims. The learned-frequency comparator appears in App. E.

B.2 REPRODUCIBILITY SNAPSHOT

The evidence ladder separates causal identification from architecture, scale, adaptation, frozen-checkpoint, and cross-modal consequences. Scratch text runs use FineWeb-Edu (Penedo et al., 2024).

Table 3: Evidence and protocol map. Every row retains its own intervention, experimental unit, and endpoint.

Role	Model/regime	Intervention	Unit	Endpoint	Appendix
Allocation identification	151.9M scratch	fixed support, 30 interiors	3 training seeds	tail NLL	C.1
Shape/configuration breadth	50.9M factorial	fixed support, 5 allocations	12 configs $\times$ 3 seeds	weighted OOD NLL	C.2
Scarce rotary budget	432M MLA-style	Geo/EVQ-COSH, $K=16$	3 training seeds	PPL	G.1
Range composition	454M MHA	same YARN-STYLE on Geo/EVQ-COSH	3 seeds/substrate	PK/PPL	G.2
Full-parameter scale	750M/1.485B	continued/from-init Geo/EVQ-COSH	one matched pair each	PPL/retrieval	B.3/F.1
Frozen allocation	OLMo/Qwen	same support and amplitude	20 rows/task	RULER	F.2
Adaptation/source use	1.485B/8B	matched LoRA/source block	one trained pair	QA/RULER/NLL	F.1/F.3
Cross-modal scope	129.6M video DiT	temporal Geo/EVQ-COSH	2 training seeds	denoising MSE	B.5

Table 4 lists the training hyperparameters shared across the supporting text experiments.

Table 4: Training hyperparameters for the supporting scratch text experiments.

Parameter	Value
Optimizer	AdamW ($\beta_1=0.9$, $\beta_2=0.95$, $\epsilon=10^{-8}$)
Learning rate	6×10^{-4} (50M), 3×10^{-4} (125M), 2×10^{-4} (454M/MLA/750M)
LR schedule	Cosine decay to $0.1 \times$ peak LR
Warmup	2% of total steps (linear)
Batch size (seqs)	32 (50M), 16 (125M), 2–4 (454M/750M), 6 (MLA)
Weight decay	0.1
Gradient clipping	max-norm 1.0
Dropout	0.0
Sequence packing	Contiguous, no cross-document masking
<i>Video DiT (129.6M)</i>	
Optimizer	AdamW ($\beta_1=0.9$, $\beta_2=0.95$)
RoPE base	10,000 (temporal axis)
Activation	GELU (tanh approx.)
<i>MLA (432M)</i>	
LR / warmup	2×10^{-4} / 2% of steps
Training tokens	500M

Table 5: 750M continued-pretraining check (seed 42; 2K $\rightarrow$ 4K; 500M continuation tokens). Passkey retrieval is teacher-forced NLL-gap and is saturated at 8K; AR exact is greedy autoregressive exact match over 40 trials.

Metric	Geo	EVQ-COSH
PPL@2K	25.9	26.2
PPL@4K	22.0	22.3
PPL@8K	23.4	19.6
PPL@16K	45.1	24.4
NLL-gap retrieval @8K	100%	100%
Strict AR exact @8K	0%	77.5%
Global strict AR exact	66.67%	92.5%

Within each text comparison, the architecture, optimizer, data order, and training loop are fixed, and the only intervention is the initialized inverse-frequency table. The scratch-text experiments use base $b=500K$ unless stated otherwise.

B.3 LARGER-SCALE SUPPORTING EVIDENCE

The 750M ‘continue@4K’ experiment starts both arms from the same 2K Geo checkpoint and applies a 500M-token continuation with either the Geo or EVQ-COSH table (seed 42). At 16K, PPL is 45.1 versus 24.4; on 40 passkey trials at 8K, strict autoregressive exact match is 0% versus 77.5%. The EVQ-COSH continuation uses midpoint quantisation with $\tau=1.5$. Both arms are full-parameter continuations from the shared checkpoint.

B.4 MODALITY-DEPENDENT τ CORRECTION FOR VIDEO DiT

The text-style reference used for this sweep gives $\tau=K_t/\sqrt{T}=2.83$ for $K_t=16$ and $T=32$; the best tested Video DiT point is $\tau=1.5$ (0.53 $\times$), which selects the supporting video configuration.

B.5 CROSS-MODAL EVIDENCE: VIDEO DiT WITH BIDIRECTIONAL ATTENTION

VideoRoPE and RIFLEx make temporal-frequency allocation explicit in video models (Wei et al., 2025; Zhao et al., 2025). We test the same allocation principle in a 129.6M video diffusion transformer with bidirectional attention and 3D RoPE on Oscillating Moving MNIST (Srivastava et al., 2015).

Training uses 32 frames; evaluation uses 128 frames ($4\times$) with temporal YARN-STYLE. For each seed, the Geo and EVQ-COSH ($\tau=1.5$) branches are trained in the same head-to-head run.

Table 6 reports two independent seeds. The within-seed contrasts favour EVQ-COSH on train, all-frame extrapolation, and far-frame extrapolation MSE in both runs; the two-seed mean deltas are -26% , -16% , and -33% , extending the scope check to bidirectional 3D RoPE.

Table 6: DiT head-to-head evaluation (129.6M, bidirectional attention, 3D RoPE). Each seed is one shared-run comparison; values are denoising MSE at $t=0.5$.

Seed	Metric (YARN-STYLE)	Geo ($\tau=0$)	EVQ-COSH ($\tau=1.5$)	Δ
42	Train MSE	0.00911	0.00720	-21%
42	All extrap	0.00724	0.00606	-16%
42	Far extrap	0.00989	0.00639	-35%
137	Train MSE	0.01113	0.00761	-32%
137	All extrap	0.00713	0.00598	-16%
137	Far extrap	0.00856	0.00601	-30%
Mean Δ		$-26\% / -16\% / -33\%$		

Low-phase-channel diagnostic. At base = 10,000, six of the 16 temporal frequencies accumulate less than 0.1 rad over the 32 training frames. In the pre-specified diagnostic, $\tau=1.2$ leaves 11/16 channels above this threshold (effective rank 9), whereas $\tau=1.5$ leaves 12/16 (rank 10). Repeating the head-to-head comparison at base=1,000 places 13/16 geometric channels above the threshold. At seed 42, $\tau=1.2$ and $\tau=1.5$ differ by less than 0.4% on the reported metrics and both improve far-frame MSE by about 48% relative to Geo, supporting the same low-phase-channel diagnostic.

Table 7: Base = 1,000 DiT head-to-head (129.6M, seed 42, YARN-STYLE). The two EVQ-COSH settings are within 0.4% on the reported metrics in this run.

τ	Train MSE	All extrap	Far extrap	$\Delta_{\text{far vs Geo}}$
0.0 (Geo)	0.02037	0.01274	0.01646	—
1.2	0.00933	0.00820	0.00859	-48%
1.5	0.00930	0.00822	0.00858	-48%

B.6 WHAT THE SCHEDULE CONCRETELY DOES

At $\tau=4$, $b=5 \times 10^5$, the warp moves budget toward short inverse-frequency scales: the continuous quantile $u=0.5$ moves from $1/\omega=707$ to 9.69, and $u=0.9$ from 134,609 to 1,720. The deployed midpoint grid therefore changes sampled extrema and span at finite K ; the exact-range experiments anchor those quantities explicitly. At $\tau=2$, $\rho_\tau(0.1)=1.71$ and $\rho_\tau(1)=0.55$, while the band $1/\omega=2048$ has relative density 0.76. Thus the finite grid thins most strongly at scales much slower than the training window while retaining more of the training-scale band. The exact-range experiments measure the corresponding trained allocation effect, and Table 1 separately measures incompatibility after a frozen post-hoc swap.

C IDENTIFICATION PROTOCOL AND PROVENANCE

C.1 EXACT-RANGE CONTROL AT 151.9M

What is pinned. The control is designed so that every scalar quantity a base/range method can express is identical between arms. Within each seed pair the two arms share: model architecture (151,898,880 parameters); trainable initialisation; training length 256; token order; optimiser, LR schedule, global batch; within-pair micro batch and accumulation; training budget (499,974,144 tokens, 7,629 optimiser steps); the 32 frozen evaluation anchors; the sampled highest frequency; the sampled lowest frequency; and the log-frequency span. The single free variable is the location of the $K-2 = 30$ interior frequencies.

The optimiser is fused AdamW with $\beta_1=0.9$, $\beta_2=0.95$, weight decay 0.01, and gradient clipping at 1.0. Peak/minimum learning rates are $6 \times 10^{-4}/6 \times 10^{-5}$ with 762 warmup steps and cosine decay. The global batch is 256 sequences of length 256 for every arm. Seed 42 uses micro batch 64 with four accumulation steps; seeds 137 and 256 use micro batch 128 with two accumulation steps. Each paired comparison therefore has identical execution geometry, optimiser steps, and counted-token budget; the aggregate uses training seed as its statistical unit under this shared scientific protocol.

Model and data. The model has 12 layers, hidden size 768, 12 heads of dimension 64, MLP width 3,072, and vocabulary size 50,304. Training uses the unshuffled FineWeb-Edu (Penedo et al., 2024) sample/10BT/000_00000.parquet shard at pinned revision 87f09149ef4734204d70ed1d046ddc9ca3f2b8f9; validation uses the disjoint sample/10BT/004_00000.parquet shard. Both are tokenised with the GPT-NeoX-20B tokenizer (Black et al., 2022) at revision c292233c833e336628618a88a648727eb3dff0a7.

Arms. *Baseline:* FMROPE at fixed training support, implementing the uniform-in-log exponent grid specified in §6.1 of Oka et al. (2026b) at training base 256. *Intervention:* anchored EVQ-COSH at $\tau=4$. The anchored and deployed midpoint EVQ-COSH tables are different support embeddings of the *same normalised shape*: both use midpoint Cosh quantiles q_k , and the diagnostic removes only their affine endpoint degrees of freedom. This normalisation is what makes the exact-range control possible.

Concretely, let $q_k = \Phi_\tau((k + \frac{1}{2})/K)$, $s_k = (q_k - q_0)/(q_{K-1} - q_0)$, and $R = ((K-1)/K) \log B$. The intervention uses

$$\omega_k^{\text{anchor}} = \exp(-Rs_k), \quad \omega_k^{\text{FMR}} = \exp\left(-R \frac{k}{K-1}\right). \quad (29)$$

Hence $k=0$, $K-1$ match exactly and only the $K-2$ interior entries differ.

Evaluation. Two conditions. *fixed*: each checkpoint retains its training range. *target-matched*: the range is set to the declared evaluation length while preserving each arm’s normalised interior spacing. Metric is paired final-128-token teacher-forced NLL over the 32 frozen anchors; reported PPL is $\exp(\text{mean NLL})$.

Three-seed contrasts. Negative values favour anchored EVQ-COSH. The aggregation unit is the training seed; evaluation anchors remain paired observations within a seed.

Table 8: Exact-range anchored EVQ-COSH minus FMROPE tail-NLL contrasts. Mean $\pm$ SD is across three independent training seeds.

Length	Seed 42	Fixed training range			Mean $\pm$ SD	Target-matched range			Mean $\pm$ SD
		Seed 137	Seed 256			Seed 42	Seed 137	Seed 256	
256	+0.033	+0.029	+0.017	+0.026 $\pm$ 0.008	+0.033	+0.029	+0.017	+0.026 $\pm$ 0.008	
512	-0.478	-0.271	-0.094	-0.281 $\pm$ 0.192	+0.061	+0.053	+0.067	+0.060 $\pm$ 0.007	
1,024	-0.205	-0.191	-0.132	-0.176 $\pm$ 0.039	+0.182	+0.156	+0.344	+0.227 $\pm$ 0.102	
2,048	-0.113	-0.181	-0.143	-0.146 $\pm$ 0.034	+0.279	+0.399	+0.701	+0.460 $\pm$ 0.218	

Under *fixed*, anchored EVQ-COSH is favoured by 3/3 seeds at every OOD length; under *target-matched*, FMROPE is favoured by 3/3 seeds at every OOD length. The fixed-range sign agrees across all seeds and OOD lengths; the 1K/2K magnitudes are the most stable across seeds.

Reading. The *fixed* column identifies the optimisation variable: interior allocation is not reducible to scalar base/range selection. Under the tested target-retargeted policy, the ordering reverses. The two columns therefore establish that the fixed-support result is conditional on the selected range; they do not identify a support-by-allocation interaction law. The *fixed* column isolates allocation when one table serves multiple lengths (§2.1).

C.2 EXACT-RANGE FACTORIAL AT 50.9M

Design. The schedule arms, including the $1.25\times$ multiplier, were specified before any result was inspected. Grid: 2 bases $\times$ 2 training lengths $\times$ 3 head dimensions = 12 structural config-

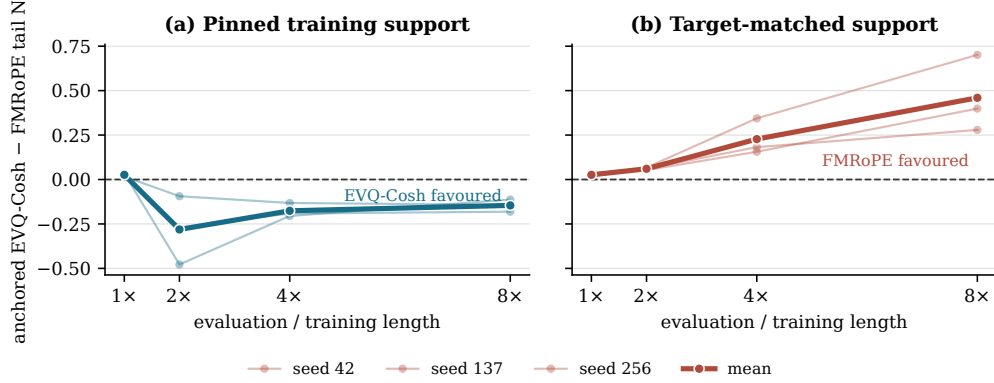


Figure 4: **Support and interior allocation are distinct design coordinates.** Each thin line is one training seed and the heavy line is the mean. At pinned training support, all three seeds favour anchored EVQ-COSH at every OOD length (a). Retargeting each arm’s support reverses all three OOD comparisons (b). The fixed-support panel isolates the allocation effect; the tested target-matched policy reverses the preferred allocation.

Table 9: **Exact-range factorial** (50.9M; 2 bases $\times$ 2 training lengths $\times$ 3 head dimensions = 12 configurations, 3 seeds). All arms share sampled extrema and log-span with Geo. Metric: weighted OOD NLL over $2 \times / 4 \times / 8 \times$, paired within configuration and seed. All arms were pre-specified. CIs are 10,000 configuration-level bootstraps; p is a two-sided exact sign-flip over all 2^{12} assignments, unadjusted across contrasts, so the claim rests on the cross-configuration direction (App. C).

Contrast	Δ NLL	95% CI	$\Delta < 0$ configs	p
EVQ-COSH $0.75 \times$ – Geo	−0.00912	[−0.018, −0.001]	8/12	0.082
EVQ-COSH rule – Geo	−0.00988	[−0.021, 0.002]	7/12	0.125
EVQ-COSH $1.25 \times$ – Geo	− 0.01210	[−0.021, −0.003]	10/12	0.027
Matched exponential – Geo	−0.01062	[−0.021, −0.001]	9/12	0.071
<i>The matched exponential reproduces the non-uniform direction:</i>				
EVQ-COSH rule – exponential	+0.00074	[−0.006, 0.008]	7/12	0.836

urations; 5 schedules (Geo, anchored EVQ-COSH at $0.75 \times$, $1.00 \times$ [the formula point], $1.25 \times$, and a deformation-matched exponential); 3 seeds. Main factorial 180/180 completed, boundary- τ follow-up 12/12 completed, covering the complete pre-specified matrix.

The exponential control uses $z_\gamma(t) = (e^{\gamma t} - 1)/(e^\gamma - 1)$ on $t \in [0, 1]$. Its γ is chosen without task results so that the RMS node displacement from the uniform grid matches the formula-Cosh displacement, while the sampled endpoints and log-span remain identical across arms.

Metric. Weighted OOD NLL over $2L/4L/8L$,

$$\bar{N}_{\text{OOD}} = \frac{\sum_{r \in \{2,4,8\}} \log_2(r+1) N_{rL}}{\sum_{r \in \{2,4,8\}} \log_2(r+1)},$$

paired within structural configuration and seed, averaged within seed, then equally weighted over the 12 configurations. The statistical unit is the configuration. Equivalent-PPL figures quoted anywhere are $\exp(\text{mean NLL})$, never an arithmetic mean of raw PPL.

Absolute values. Mean weighted OOD NLL (equivalent PPL): Geo 6.211973 (498.68); EVQ-COSH $0.75 \times$ 6.202858 (494.16); EVQ-COSH $1.00 \times$ 6.202094 (493.78); EVQ-COSH $1.25 \times$ 6.199873 (492.69); matched exponential 6.201354 (493.42).

Boundary- τ arms. At formula $\tau=1$ the $1.5 \times$ arm is better; at formula $\tau=8$ the $0.75 \times$ arm is. The improving direction points inward at both tested extremes; these discrete arms do not locate continuous basin boundaries or yield a replacement closed-form rule.

Scope. This is a short-budget 50.9M mechanistic NLL control trained on a deterministically repeated WikiText-2 raw stream (Merity et al., 2017), a different corpus from the 151.9M control and from the mature-scale runs. The factorial establishes cross-configuration *direction*, while the 151.9M control provides the effect size within its configuration. Scale is evaluated in §5.1.

D THE YARN-STYLE OPERATOR

Every arm marked +YARN-STYLE uses the repository’s fixed-index operator, inspired by the NTK-by-parts/YaRN family (Peng et al., 2024): high-frequency channels are left nearly unchanged and lower-frequency channels receive progressively more range scaling. The implementation, cut indices, and scale s are held fixed across the compared arms. Because the underlying frequencies differ, the realised phase movement is not matched channel by channel. The result is therefore a composition test of the same index-space rule on two training-time tables, not a movement-matched mechanism control.

Implemented components. For K frequency channels, the scaler uses fixed channel-index cut points $k_{\text{lo}} = \lfloor 0.20K \rfloor$ and $k_{\text{hi}} = \lfloor 0.90K \rfloor$. Between them it applies the cubic smoothstep $r(x) = x^2(3 - 2x)$ and replaces each inverse frequency ω_k by

$$\omega_k / \left(s^{r_k} t^{r_k/2} \right), \quad t = 1 + 0.07 \log_2 s.$$

Its boundaries are fixed by channel index; the factor t is folded into frequency scaling. These implementation choices define YARN-STYLE and distinguish it from the cited YaRN reference method.

E LEARNED-FREQUENCY COMPARATOR

Table 10: Learned-frequency comparator (125M, $L_{\text{train}}=128$, FineWeb-Edu, $128 \rightarrow 8K$). Geo, learned-inv-freq, and EVQ-COSH are seed 42; learnable- τ is 3-seed (42/137/256). Allocation-shape attribution uses the fixed, zero-parameter analytic schedules of §2.1; this separately reported arm is a layer-shared learnable inverse-frequency baseline — one learned scalar per band, shared across layers and heads — and therefore belongs to the learned-*table* family (Karypis et al., 2026); DAPE belongs to the learned positional-*operator* family (Zheng et al., 2024).

Method (128 $\rightarrow$ 8K)	Extra params	PPL@128	PPL@8K	Δ vs Geo
Geo	0	184.9	513.7	—
Learnable τ	1	181.2 $\pm$ 1.3	437.9 $\pm$ 12.2	−14.8%
Learned inv-freq (100 $\times$ positional LR)	32	183.6	455.3	−11.4%
same, 10 $\times$ positional LR	32	—	477.7	−7.0%
EVQ-COSH	0	182.0	333.7	−35.0%

The 32-parameter arm in Table 10 is a layer-shared learnable inverse-frequency baseline: one learned scalar per frequency band, shared across layers and heads. It received a dedicated positional-parameter learning-rate sweep at 10 $\times$ and 100 $\times$ the base rate, and the stronger 100 $\times$ setting is the 455.3 row. It answers the learned-table tuning question. Allocation-shape attribution uses the fixed analytic schedules under an identical zero-parameter operator in §2.1.

On method identity: DAPE (Zheng et al., 2024) is a learned positional *operator*, whereas this comparator learns inverse frequencies directly and belongs to the learned-*table* family later studied at scale by Karypis et al. (2026). In this protocol, the learned table improves over Geo but remains weaker than EVQ-COSH; the comparison does not identify whether the gap comes from the final table, optimization dynamics, or the positional-parameter learning rate.

Table 11: **RULER 13-family scores (%)**. Within each protocol, continuation data and evaluation are matched and one seed is used per arm; evaluation uses 20 examples per family and length. OLMo Q/K-only freezes inherited V/O LoRA and trains only the tensors RoPE acts on; LLaMA uses matched Q/K/V/O LoRA. Training and evaluation rows are disjoint and share the 13 generator families, measuring task-family-adapted length transfer.

Model / protocol	Arm / metric	1 ×	2 ×	4 ×
OLMo-2 1.485B, full Q/K/V/O (4K cap)	Native, official-scorer macro	82.16	0.08	0
	EVQ-COSH, official-scorer macro	37.51	21.29	6.13
OLMo-2 1.485B, Q/K-only (4K cap)	Native, official-scorer macro	72.19	2.02	0.38
	EVQ-COSH, official-scorer macro	42.44	31.63	5.03
LLaMA-3-8B, Q/K/V/O (8K cap)	Native, official-scorer macro	94.44	0.295	—
	EVQ-COSH, official-scorer macro	77.60	14.03	—
	Native, normalized exact	17.69	0	—
	EVQ-COSH, normalized exact	21.54	1.54	—

F MATURE-SCALE PROTOCOLS

F.1 OLMo-2 1.485B

All OLMo-2 EVQ-COSH arms use the endpoint grid $u_k=k/K$ with $\tau=2$ and base 500,000; the frequency tensor is fixed throughout each continuation.

Matched routing conversion. We use OLMo-2-0425-1B-Instruct (OLMo Team, 2025) (1.485B actual parameters) with rank-64, alpha-128 Q/K/V/O LoRA (Hu et al., 2022). Native and EVQ-COSH share the checkpoint, training rows and order, token budget, optimizer, and evaluation rows; the fixed frequency substrate is the intended intervention. Every backward pass contains at most 4K physical tokens, but explicit position IDs expose target-range relative phases. This is therefore a physical-memory cap, not phase-free 4K training. On 8K `niah_single_1`, strict first-generated-number exact is 0/100 for Native and 69/100 for the matched EVQ-COSH seed; an independently trained EVQ-COSH seed scores 67/100 on the same rows. The two EVQ-COSH results are training-seed replications on a shared evaluation set. Natural-text NLL and the distinct autoregressive endpoints are kept separate in the discussion below and in Table 11.

Selective Q/K phase adaptation. To test a less intrusive mature-checkpoint adaptation, we start from the protocol-matched Native and EVQ-COSH Stage-A parents, freeze every inherited V/O LoRA tensor within each arm (verified bitwise), and continue only the existing Q/K tensors for 300 steps. Each arm trains 8,388,608 Q/K parameters while freezing the 8,388,608 inherited V/O parameters, with global batch 8, learning rate 5×10^{-5} , and 20 warmup steps. The QA and RULER generators use seeds 20260728 and 20260729, respectively; the bundled `Q/K metrics.json` records the full protocol. Both arms use the same 1:1:2 mixture of continuous-4K, target-8K, and target-16K phase exposure, complete answer-plus-EOS supervision, and physical sequences no longer than 4K; long phases again use explicit position IDs. On 200 held-out 2WikiMultiHopQA rows per length (Ho et al., 2020), Native/EVQ-COSH token-F1 is 25.99/24.84% at 4K, 0.07/21.48% at 8K, and 0/8.57% at 16K; exact match is 22.0/21.5%, 0/17.5%, and 0/4.0%, respectively. Prompts retain the official question and context and are deterministically filled with answer-filtered 2Wiki distractors to reach the target length, forming a held-out 2Wiki length-transfer evaluation. The RULER adapters are independent of the QA adapters: Table 11 reports both the earlier 276-step full-Q/K/V/O continuation and the 300-step selective-Q/K continuation, each trained over all 13 families. Evaluation uses 20 examples per family and length. A paired family-stratified bootstrap over the tracked Q/K-only per-example predictions gives EVQ-COSH-minus-Native official-scorer macro differences of -29.75 points at 4K (95% CI $[-34.42, -24.85]$), $+29.61$ at 8K ($[26.54, 32.74]$), and $+4.65$ at 16K ($[3.17, 6.38]$). These intervals condition on the single trained pair and quantify evaluation-row uncertainty, not training-seed variability.

From initialisation. Starting from the public step-0 initialisation, the same-recipe Geo/EVQ-COSH comparison gives PPL 177.99/191.36, 161.19/167.45, 163.88/156.87, and 182.73/159.64

at 2/4/8/16K under a matched 2.097B-token budget. The EVQ-COSH-minus-Geo NLL is $+0.0724/+0.0381/-0.0437/-0.1351$ at 2/4/8/16K, with 122/128 and 126/128 held-out documents favouring EVQ-COSH at 8K and 16K. This is one same-initialisation, same-scientific-recipe natural-text LM comparison; capability is evaluated by the separate autoregressive protocols above.

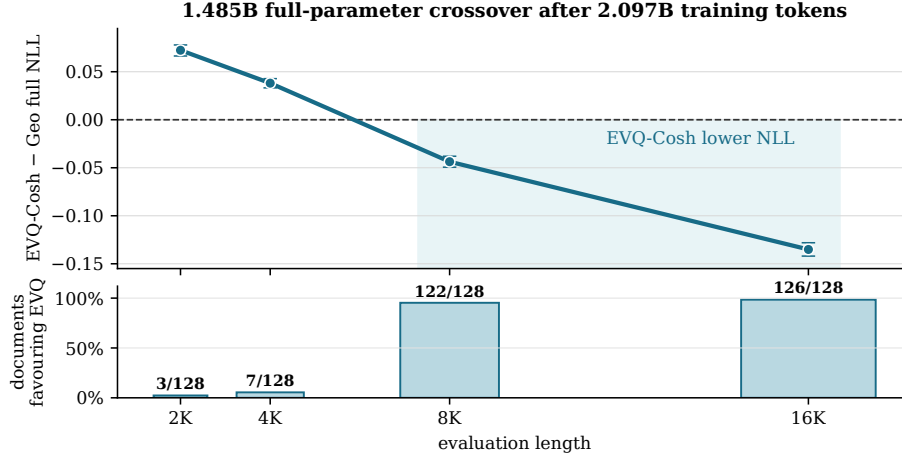


Figure 5: **Full-parameter scale evidence crosses beyond the training window.** The 1.485B Geo and EVQ-COSH arms start from the same public initialisation and follow the same scientific recipe for 2.097B tokens. Error bars are paired bootstrap intervals over the same 128 held-out documents, conditioned on this trained pair; the lower panel gives the document count favouring EVQ-COSH at each length.

F.2 FROZEN FIXED-SUPPORT ALLOCATION

Table 12: **Frozen fixed-support intervention.** Macro score (%); higher is better. Each row is a separate fixed-checkpoint comparison on its stated RULER task set.

Checkpoint	Endpoint	Geometric	Movement ramp	Frozen derived
OLMo-2 1.485B	unseen-nine, 16K	0.56	61.04	60.47
Qwen2.5-1.5B	core-four, 64K	57.75	64.00	66.50

The mature-checkpoint intervention in Table 12 holds both models frozen. For each checkpoint, all three tables contain 64 rotary pairs and share the Native fast endpoint and the Native slow endpoint divided by four. They also share the factor-four attention amplitude $1 + 0.1 \ln 4$, evaluation rows, greedy decoder, precision, and hardware. The same-support geometric arm is log-linear between the endpoints. The nearest movement-profile ramp is the discrete YaRN-family linear ramp minimising movement-profile MSE to the frozen derived table; its choice uses no task labels. The remaining arm installs the frozen derived interior allocation.

OLMo-2-0425-1B-Instruct uses nine confirmation-only RULER tasks at 16K, 20 rows per task. Qwen2.5-1.5B-Instruct (Yang et al., 2024) uses the four development RULER tasks at 64K, also with 20 rows per task; it evaluates a factor-four table at twice the Native context length. Row resampling conditions on each fixed checkpoint and task set. Both non-geometric allocations produce the large OLMo recovery. The derived-minus-geometric intervals are $[54.88, 64.80]$ points for OLMo and $[0.25, 17.50]$ points for Qwen; the paired derived-minus-ramp intervals contain zero on both models. The control therefore attributes the effect to fixed-support reallocation while leaving profile-specific ordering unresolved.

The complete OLMo deployment policy is a separate, bundled intervention: it uses exact Native when observed prefill plus generation reserve fits the Native window and otherwise chooses one frozen factor-four table and amplitude before prefill, without changing the route during the KV-cache lifetime. It adds no learned parameters or training tokens. On a post-policy-freeze, full-200 Qasper

evaluation (Dasigi et al., 2021) at 16K, this policy obtains token F1 24.57% versus 18.03% for official Transformers YaRN factor four. Both arms use identical rows, and every post-chat input plus its 128-token generation reserve fits within 16K. This row evaluates the complete deployment policy on natural context; Table 12 provides the allocation-only contrast.

F.3 LLAMA-3-8B-INSTRUCT

The EVQ-COSH arm in both LLaMA protocols uses midpoint quantisation with $\tau=1.414$; Native retains the pretrained model’s original frequency table. The training objectives, adapters, and endpoints remain separate below.

Probability and causal source use. In the matched seed-42, 300-step Q/K/V/O LoRA protocol, EVQ-COSH-minus-Native NLL is $+0.390/-1.510/-2.048$ at 8/16/32K. On ten frozen true-16K passkey cases, median target-block hit@16 over 32 retrieval heads rises from 18.75% to 64.06%. Blocking answer-side decode attention to the remote gold block in every head preserves all prompt tokens, original rotary KV indices, and the 128-token block geometry; it changes NLL by -0.0095 for Native and $+1.5055$ for EVQ-COSH. These are matched teacher-forced probability and position-preserving causal-routing measurements; autoregressive task accuracy is evaluated separately. Descriptive Native/EVQ-COSH PPL is 6.817/10.068, 108.958/24.068, and 991.475/127.911 at those lengths.

RULER-family continuation. A separate 516-step continuation uses identical physical-8K supervision over all 13 RULER families, with one seed per arm and disjoint generated training/evaluation rows. Evaluation uses 20 examples per family and length. At 16K, Native/EVQ-COSH official-scorer macro is 0.295/14.03% and normalized exact is 0/1.54%. At 8K, Table 11 reports official scorer macro (94.44/77.60%) and normalized exact (17.69/21.54%).

Endpoint scope. NLL, attention routing, strict autoregressive exact match, multi-hop QA, and RULER remain separate endpoints. The 2Wiki and RULER continuations use the same task families at train and test with disjoint rows, establishing task-family-adapted length transfer. The natural-text and routing rows evaluate probability and causal source use.

G SUPPORTING RESULTS

G.1 MULTI-HEAD LATENT ATTENTION

We test the finite-budget hypothesis in a 432M-parameter transformer following the MLA architectural pattern (DeepSeek-AI, 2024), with 24 layers, width 1024, 16 heads, and per-head width 64. Its decoupled rotary subspace has $d_{\text{rope}}=32$ ($K=16$ frequencies); the remaining $d_{\text{nope}}=32$ dimensions are non-rotary, and the shared KV latent rank is 256. The model trains from scratch on 500M FineWeb-Edu tokens (Penedo et al., 2024) at $L_{\text{train}}=8192$ and base 500K over seeds 42, 43, and 88. EVQ-COSH uses the pre-specified empirical $\tau=1.414$ setting. Because the compressed rotary subspace does not instantiate the full-RoPE $d_{\text{eff}}=d_{\text{head}}$ convention in Eq. (8), this experiment tests frequency allocation at fixed architecture rather than the operating rule.

For the supporting overlay, let $\lambda_k = 2\pi/\omega_k$ and $t_k = \text{clip}((\lambda_k - L_{\text{train}})/(31L_{\text{train}}), 0, 1)$. The MLA wavelength-blend operator sets $\omega'_k = \omega_k/[1 + t_k(s - 1)]$: it preserves wavelengths below the training length, applies the full factor s above $32L_{\text{train}}$, and interpolates between them. This run-specific operator is distinct from YARN-STYLE.

EVQ-COSH reduces 16K PPL by 31.1% ($138.8 \rightarrow 95.6$), while 8K PPL changes by $+0.9\%$. All three training seeds favour EVQ-COSH at 16K. The EVQ-COSH plus MLA wavelength-blend row has the lowest mean PPL at 16K and beyond. These rows establish the tested ordering for this architecture and protocol.

Table 13: MLA validation (432M, $d_{\text{rope}}=32$, 3-seed mean $\pm$ std). Rows marked “MLA wavelength-blend” use this run’s wavelength-blend operator, which is distinct from YARN-STYLE.

Method	PPL@8K	PPL@16K	PPL@24K	PPL@32K
Geo	35.4 $\pm$ 0.9	138.8 $\pm$ 5.5	241.5 $\pm$ 2.6	323.7 $\pm$ 5.2
Geo + MLA wavelength-blend ($s = 4$)	35.5 $\pm$ 0.9	117.9 $\pm$ 6.5	204.9 $\pm$ 6.9	278.5 $\pm$ 6.3
EVQ-COSH	35.8 $\pm$ 0.8	95.6$\pm$4.1	204.9 $\pm$ 14.7	291.6 $\pm$ 20.5
EVQ-COSH + MLA wavelength-blend ($s = 4$)	35.8 $\pm$ 0.8	71.1$\pm$4.1	153.2$\pm$12.5	236.6$\pm$15.9

Table 14: **Substrate $\times$ range composition** (454M, $L_{\text{train}}=2048$, 10% passkey mix, 3 seeds). YARN-STYLE is the repository’s fixed-index operator (App. D), applied at the *same* fixed scale $s=8$ to both substrates. PK is teacher-forced NLL-gap retrieval; PK@8K is mean $\pm$ std, other entries are 3-seed means.

Method	PK@8K	PK@12K	PK@16K	PPL@8K	PPL@16K
Geo	41 $\pm$ 5%	57%	51%	161.9	253.2
Geo+YARN-STYLE	61 $\pm$ 3%	59%	51%	82.9	157.7
EVQ-COSH	53 $\pm$ 8%	63%	50%	150.3	229.5
EVQ-COSH+YARN-STYLE	100$\pm$0%	79%	68%	70.9	107.5

G.2 COMPOSITION WITH YARN-STYLE

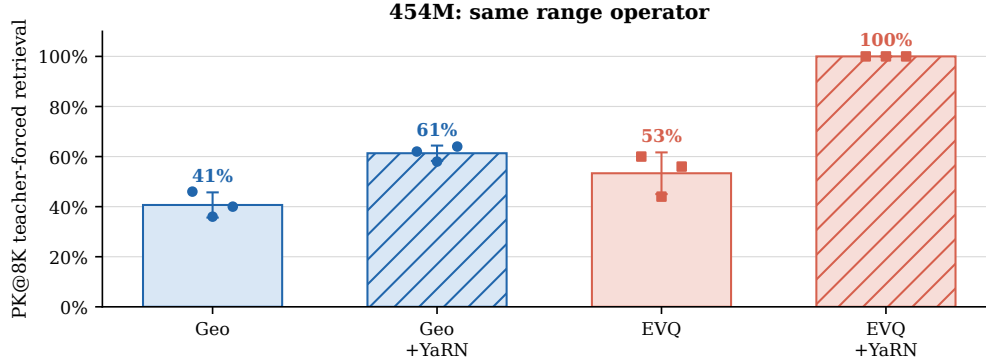


Figure 6: **The same range operator has substrate-dependent leverage in the tested protocol.** Three-seed PK@8K means, individual seeds, and standard deviations: YARN-STYLE raises Geo from 41% to 61% and EVQ-COSH from 53% to 100%.

Applying the same YARN-STYLE operator adds 20 points on Geo and 47 points on EVQ-COSH, a 27-point leverage difference under the shared code path, cut indices, scale, and evaluation rows.

Table 15: $L=256$ YARN-STYLE composition (125M, 3-seed mean PPL). The tested $\tau=4.0$ setting has the lowest mean at every reported ratio.

Method ($L=256$)	4 $\times$	8 $\times$	16 $\times$	32 $\times$
Geo+YARN-STYLE	60.7	97.5	190.4	260.2
EVQ-COSH ($\tau=2$)+YARN-STYLE	51.6	66.7	107.9	141.7
EVQ-COSH ($\tau=4$)+YARN-STYLE	50.9	60.9	84.1	99.6

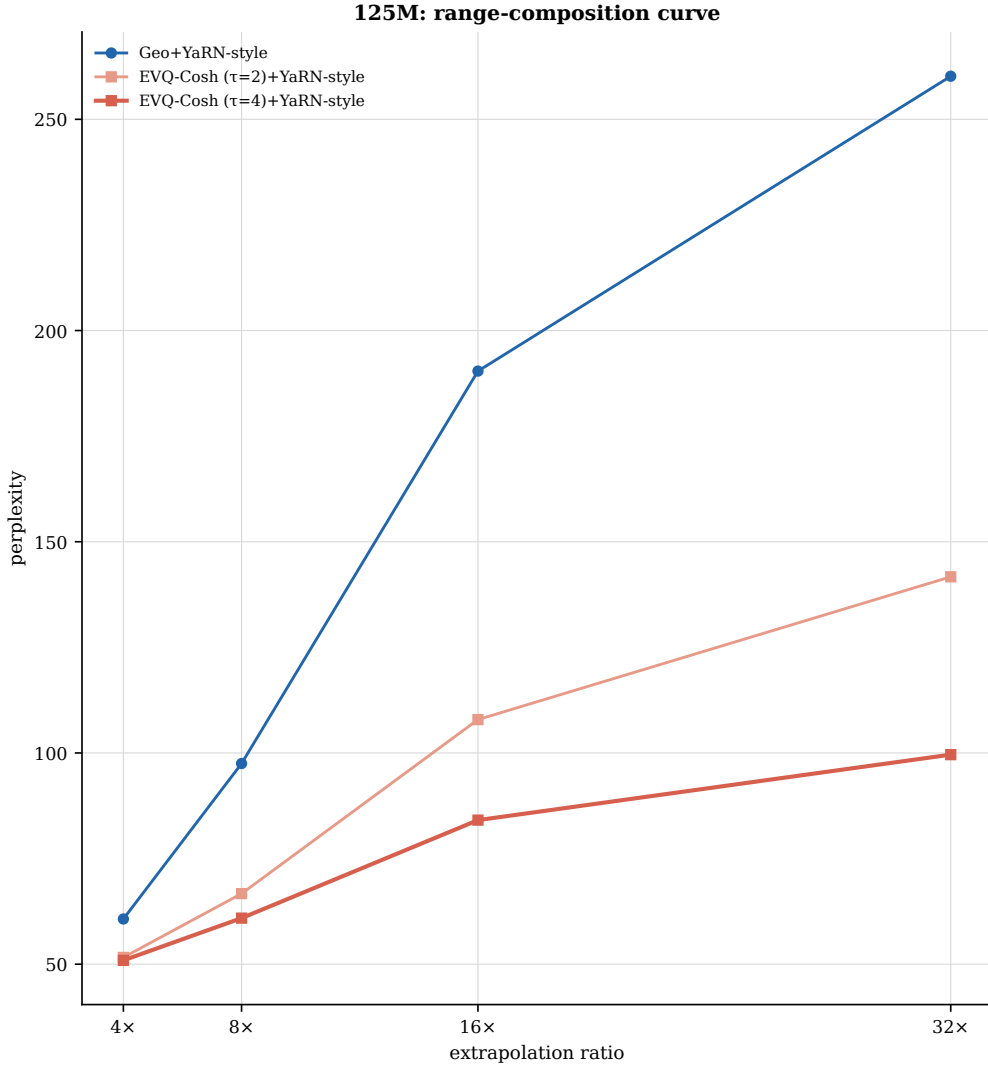


Figure 7: **The composition gap widens with extrapolation.** At 125M, the tested $\tau=4$ allocation remains lowest from $4\times$ to $32\times$ under the same YARN-STYLE rule.

Reading the composition. The tested $\tau=4$ composition lowers PPL relative to Geo+YARN-STYLE by 16%/38%/56%/62% at $4\times/8\times/16\times/32\times$. In this protocol, the same index-space rule produces different composition effects on the two learned spectral substrates.